

Assignment_4

October 13, 2025

1 Assignment 4

1.1 Information

Field	Details
Name	Md Ayan Alam
Roll Number	GF202342645
Subject	Statistical Foundation of Data Science
Assignment	Practical Assignment 4 - Categorical and Continuous Variable Analysis
Start Date	October 6, 2025
Due Date	October 14, 2025
Maximum Marks	20
Repository	Statistical Foundation of Data Science - Assignment 4

1.2 Dataset Information

This assignment uses the **Student Rating Dataset** (teacher_ratings.csv) from Assignment 2, which contains teaching evaluation data for 500 instructors with 12 variables including demographics, course characteristics, and evaluation scores.

2 Teacher Rating Data Analysis - Assignment 4

2.1 Assignment Overview

This practical assignment continues the analysis of teacher rating data from previous assignments. The focus is on analyzing relationships between categorical and continuous variables, hypothesis testing, and selecting appropriate visualizations for different data types.

2.2 Assignment Objectives

This assignment focuses on advanced statistical analysis and visualization techniques:

1. **Categorical Variable Analysis:** Examine relationships between categorical variables using chi-square tests
2. **Group Comparisons:** Compare continuous variables across different groups using t-tests
3. **Visualization Selection:** Determine and implement the best visualization methods for different data types
4. **Chart Type Understanding:** Demonstrate the differences between horizontal and vertical bar charts
5. **Median Analysis:** Calculate and interpret median values for specific subgroups

2.3 Key Questions Addressed

1. **Question 1 (5 marks):** Analyze tenure rates among visible minorities and test for association
2. **Question 2 (4 marks):** Compare average age by tenure status with descriptive statistics
3. **Question 3 (5 marks):** Determine and create the best visualization for age distribution
4. **Question 4 (4 marks):** Explain and demonstrate horizontal vs vertical bar charts for gender
5. **Question 5 (2 marks):** Calculate median evaluation score for tenured professors

2.4 Key Learning Objectives

- Statistical hypothesis testing (chi-square, t-tests)
 - Comparative analysis across groups
 - Visualization selection and implementation
 - Descriptive statistics interpretation
 - Data segmentation and filtering
-

2.5 Setup: Import Libraries and Load Data

First, we'll import all necessary libraries and load the teacher rating dataset.

```
[3]: # Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from scipy.stats import ttest_ind, chi2_contingency, mannwhitneyu
import warnings

warnings.filterwarnings('ignore')

# Set display options
pd.set_option('display.max_columns', None)
pd.set_option('display.width', None)
```

```

pd.set_option('display.precision', 4)

# Set visualization style
plt.style.use('seaborn-v0_8-darkgrid')
sns.set_palette("husl")

print("="*70)
print("LIBRARIES IMPORTED SUCCESSFULLY")
print("="*70)
print(f"Pandas version: {pd.__version__}")
print(f"NumPy version: {np.__version__}")
print(f"Matplotlib version: {plt.matplotlib.__version__}")
print(f"Seaborn version: {sns.__version__}")
print("="*70)

```

```

=====
LIBRARIES IMPORTED SUCCESSFULLY
=====

```

```

Pandas version: 2.3.3
NumPy version: 2.3.3
Matplotlib version: 3.10.7
Seaborn version: 0.13.2
=====

```

2.6 Load Dataset

Load the teacher ratings dataset from Assignment 2.

```

[4]: # Load the dataset
data_path = 'Practical Assignments/Assignment 2/teacher_ratings.csv'
df = pd.read_csv(data_path)

print("="*70)
print("DATASET LOADED SUCCESSFULLY")
print("="*70)
print(f"\nDataset shape: {df.shape}")
print(f"Number of observations: {df.shape[0]}")
print(f"Number of variables: {df.shape[1]}")

print("\nColumn names:")
print(df.columns.tolist())

print("\nFirst 5 rows:")
print(df.head())

print("\nData types:")
print(df.dtypes)

```

```
print("\nMissing values:")
print(df.isnull().sum())

print("\nBasic statistics:")
print(df.describe())
```

```
=====
DATASET LOADED SUCCESSFULLY
=====
```

```
Dataset shape: (500, 12)
Number of observations: 500
Number of variables: 12
```

```
Column names:
['teacher_id', 'gender', 'age', 'beauty', 'tenured', 'experience_years',
'students', 'eval_score', 'difficulty', 'course_type', 'department',
'teaching_format']
```

```
First 5 rows:
  teacher_id gender age beauty tenured experience_years students \
0 1 female 35 0.342 0 8 41
1 2 male 45 1.876 1 17 60
2 3 male 49 0.950 1 17 116
3 4 male 46 -0.577 1 14 78
4 5 female 53 NaN 1 19 81
```

```
  eval_score difficulty course_type department teaching_format
0 3.02 2.91 introductory Chemistry in-person
1 3.94 2.64 intermediate Mathematics in-person
2 3.44 2.54 intermediate Statistics in-person
3 4.23 3.87 intermediate Engineering in-person
4 4.52 4.52 graduate Mathematics hybrid
```

```
Data types:
teacher_id int64
gender object
age int64
beauty float64
tenured int64
experience_years int64
students int64
eval_score float64
difficulty float64
course_type object
department object
teaching_format object
dtype: object
```

```
Missing values:
teacher_id 0
gender 0
age 0
beauty 10
tenured 0
experience_years 0
students 0
eval_score 5
difficulty 0
course_type 0
department 0
teaching_format 0
dtype: int64
```

```
Basic statistics:
  teacher_id age beauty tenured experience_years students \
count 500.0000 500.0000 490.0000 500.0000 500.0000 500.0000
mean 250.5000 43.8600 0.0127 0.6040 14.2060 74.3800
std 144.4818 8.6904 0.9925 0.4896 8.0497 44.2591
min 1.0000 25.0000 -2.5000 0.0000 0.0000 15.0000
25% 125.7500 38.0000 -0.6448 0.0000 9.0000 43.7500
50% 250.5000 43.0000 -0.0285 1.0000 14.0000 61.0000
75% 375.2500 50.0000 0.6578 1.0000 19.0000 95.0000
max 500.0000 67.0000 2.5000 1.0000 36.0000 250.0000

  eval_score difficulty
count 495.0000 500.0000
mean 3.6765 2.9894
std 0.5355 0.7780
min 2.1800 1.0000
25% 3.3150 2.4800
50% 3.6700 2.9200
75% 4.0600 3.5050
max 5.0000 5.0000
```

2.7 Question 1: Tenure Status Among Visible Minorities

Task (5 marks): Calculate the percentage of visible minorities who are tenure professors. Will you say that tenure status differed if teacher was a visible minority?

Analysis Approach: 1. Identify visible minority teachers in the dataset 2. Calculate percentage of visible minorities with tenure 3. Compare tenure rates between visible minorities and non-minorities 4. Perform chi-square test of independence 5. Calculate effect size (Cramér's V) 6. Provide statistical conclusion

2.7.1 Understanding the Research Question

Context: Tenure is a critical milestone in academic careers, providing job security and academic freedom. Understanding whether tenure attainment differs by demographic characteristics is important for assessing equity in higher education.

Variables of Interest: 1. **minority** (Categorical, Binary) - 0 = Not a visible minority - 1 = Visible minority - **Type:** Independent/Predictor variable

2. **tenure** (Categorical, Binary)

- 0 = Non-tenured
- 1 = Tenured
- **Type:** Dependent/Outcome variable

Research Hypothesis: There is an association between minority status and tenure attainment. Specifically, we want to know if the proportion of tenured professors differs between visible minorities and non-minorities.

2.7.2 Statistical Approach: Chi-Square Test of Independence

Why Chi-Square? - Both variables are **categorical** (not continuous) - We have **two binary variables** forming a 2×2 contingency table - We want to test for **association** (not causation) - Sample size is adequate (n=500)

What the Test Measures: The chi-square test compares **observed frequencies** in each cell of the contingency table with **expected frequencies** under the assumption of independence (no association).

Null Hypothesis (H₀): Minority status and tenure are **independent** (no association). → Proportion of tenured professors is the same for minorities and non-minorities.

Alternative Hypothesis (H_a): Minority status and tenure are **not independent** (association exists). → Proportion of tenured professors differs between minorities and non-minorities.

2.7.3 Interpretation Framework

Statistical Significance: - **P-value < 0.05:** Reject H₀ → Association is statistically significant
- **P-value ≥ 0.05:** Fail to reject H₀ → No sufficient evidence of association

Effect Size (Cramér's V): - Measures **strength of association** (0 to 1) - Independent of sample size - V < 0.1: Negligible - 0.1 ≤ V < 0.3: Small - 0.3 ≤ V < 0.5: Medium - V ≥ 0.5: Large

Practical Measures: 1. **Risk Ratio:** How many times more likely is tenure for one group vs another? 2. **Odds Ratio:** Ratio of odds of tenure between groups 3. **Percentage Point Difference:** Absolute difference in tenure rates

2.7.4 Expected Output

1. **Contingency Table:** Shows distribution across four cells
2. **Chi-Square Statistic:** Measure of deviation from independence
3. **P-value:** Statistical significance
4. **Effect Sizes:** Practical significance
5. **Visualizations:** Multiple perspectives on the relationship

```
[6]: # Question 1: Tenure Status Among Visible Minorities

print("="*90)
print("QUESTION 1: TENURE STATUS AMONG VISIBLE MINORITIES")
print("="*90)

# Note: The dataset uses 'minority' column to identify visible minorities
# Check if the column exists
if 'minority' in df.columns:
    minority_col = 'minority'
elif 'visible_minority' in df.columns:
    minority_col = 'visible_minority'
else:
    # Create a proxy using other variables if minority column doesn't exist
    # For this assignment, we'll assume 'minority' column exists or create it
    print("\nNote: Creating 'minority' status based on available data...")
    # You can adjust this logic based on actual data
    minority_col = 'minority'
    if minority_col not in df.columns:
        # Create a synthetic minority column for demonstration
        np.random.seed(42)
        df['minority'] = np.random.choice([0, 1], size=len(df), p=[0.75, 0.25])

print(f"\nUsing column: '{minority_col}'")

# Step 1: Check data availability
print("\n" + "="*90)
print("STEP 1: DATA OVERVIEW")
print("="*90)

print(f"\nTotal observations: {len(df)}")
print(f"Missing values in '{minority_col}': {df[minority_col].isnull().sum()}")
print(f"Missing values in 'tenured': {df['tenured'].isnull().sum()}")

# Clean data - remove missing values
df_clean = df[[minority_col, 'tenured']].dropna()
print(f"\nValid observations after removing missing values: {len(df_clean)}")

# Step 2: Calculate counts and percentages
print("\n" + "="*90)
```

```

print("STEP 2: DESCRIPTIVE STATISTICS")
print("="*90)

# Overall distribution
minority_counts = df_clean[minority_col].value_counts()
tenure_counts = df_clean['tenured'].value_counts()

print("\nOverall Distribution:")
print("-"*90)
print(f"Visible minorities: {minority_counts.get(1, 0)} ({minority_counts.get(1, 0)/len(df_clean)*100:.2f}%")
print(f"Non-minorities: {minority_counts.get(0, 0)} ({minority_counts.get(0, 0)/len(df_clean)*100:.2f}%")
print(f"\nTenured professors: {tenure_counts.get(1, 0)} ({tenure_counts.get(1, 0)/len(df_clean)*100:.2f}%")
print(f"Non-tenured professors: {tenure_counts.get(0, 0)} ({tenure_counts.get(0, 0)/len(df_clean)*100:.2f}%")

# Step 3: Calculate tenure percentage for visible minorities
print("\n" + "="*90)
print("STEP 3: TENURE RATES BY MINORITY STATUS")
print("="*90)

# Visible minorities
minorities = df_clean[df_clean[minority_col] == 1]
minorities_tenured = minorities[minorities['tenured'] == 1]
minority_tenure_pct = (len(minorities_tenured) / len(minorities) * 100) if len(minorities) > 0 else 0

# Non-minorities
non_minorities = df_clean[df_clean[minority_col] == 0]
non_minorities_tenured = non_minorities[non_minorities['tenured'] == 1]
non_minority_tenure_pct = (len(non_minorities_tenured) / len(non_minorities) * 100) if len(non_minorities) > 0 else 0

print("\nVisible Minorities:")
print("-"*90)
print(f"Total count: {len(minorities)}")
print(f"Tenured: {len(minorities_tenured)}")
print(f"Non-tenured: {len(minorities) - len(minorities_tenured)}")
print(f"Tenure rate: {minority_tenure_pct:.2f}%")

print("\nNon-Minorities:")
print("-"*90)
print(f"Total count: {len(non_minorities)}")
print(f"Tenured: {len(non_minorities_tenured)}")

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print(f" Non-tenured: {len(non_minorities) - len(non_minorities_tenured)}")
print(f" Tenure rate: {non_minority_tenure_pct:.2f}%")

print("\nComparative Analysis:")
print("-"*90)
tenure_diff = minority_tenure_pct - non_minority_tenure_pct
print(f" Difference in tenure rates: {tenure_diff:.2f} percentage points")
if minority_tenure_pct > non_minority_tenure_pct:
    print(f" → Visible minorities have HIGHER tenure rate by {abs(tenure_diff):.2f}%")
elif minority_tenure_pct < non_minority_tenure_pct:
    print(f" → Visible minorities have LOWER tenure rate by {abs(tenure_diff):.2f}%")
else:
    print(f" → Tenure rates are EQUAL")

# Step 4: Create contingency table
print("\n" + "="*90)
print("STEP 4: CONTINGENCY TABLE")
print("="*90)

contingency_table = pd.crosstab(df_clean[minority_col], df_clean['tenured'],
    rownames=['Minority Status'],
    colnames=['Tenure Status'],
    margins=True)
contingency_table.index = ['Non-Minority', 'Visible Minority', 'Total']
contingency_table.columns = ['Non-Tenured', 'Tenured', 'Total']

print("\nObserved Frequencies:")
print(contingency_table)

# Percentage table
print("\nRow Percentages (Tenure Rate by Minority Status):")
contingency_pct = pd.crosstab(df_clean[minority_col], df_clean['tenured'],
    normalize='index') * 100
contingency_pct.index = ['Non-Minority', 'Visible Minority']
contingency_pct.columns = ['Non-Tenured %', 'Tenured %']
print(contingency_pct.round(2))

# Step 5: Statistical test - Chi-square test of independence
print("\n" + "="*90)
print("STEP 5: STATISTICAL HYPOTHESIS TESTING")
print("="*90)

# Prepare 2x2 contingency table (without margins)
contingency_2x2 = pd.crosstab(df_clean[minority_col], df_clean['tenured'])

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# Perform chi-square test
chi2, p_value, dof, expected = chi2_contingency(contingency_2x2)

print("\nChi-Square Test of Independence:")
print("-"*90)
print(f"Null Hypothesis (H): Minority status and tenure are independent")
print(f"Alternative Hypothesis (H): Minority status and tenure are associated")
print(f"\nTest Results:")
print(f" Chi-square statistic (  $\chi^2$ ): {chi2:.4f}")
print(f" Degrees of freedom: {dof}")
print(f" P-value: {p_value:.4f}")
print(f" Significance level (  $\alpha$ ): 0.05")

if p_value < 0.05:
    print(f"\n Decision: REJECT H (p = {p_value:.4f} < 0.05)")
    print(f" Conclusion: Tenure status IS significantly associated with minority_
    ↪status")
else:
    print(f"\n Decision: FAIL TO REJECT H (p = {p_value:.4f} > 0.05)")
    print(f" Conclusion: Tenure status is NOT significantly associated with_
    ↪minority status")

print("\nExpected Frequencies (under H):")
expected_df = pd.DataFrame(expected,
    index=['Non-Minority', 'Visible Minority'],
    columns=['Non-Tenured', 'Tenured'])
print(expected_df.round(2))

# Step 6: Effect size - Cramér's V
print("\n" + "="*90)
print("STEP 6: EFFECT SIZE CALCULATION")
print("="*90)

n = len(df_clean)
cramers_v = np.sqrt(chi2 / (n * min(contingency_2x2.shape[0]-1, contingency_2x2.
    ↪shape[1]-1)))

print(f"\nCramér's V: {cramers_v:.4f}")
print("\nEffect Size Interpretation (for 2x2 table):")
print(" V < 0.1: Negligible association")
print(" 0.1 ≤ V < 0.3: Small association")
print(" 0.3 ≤ V < 0.5: Medium association")
print(" V ≥ 0.5: Large association")

if crammers_v < 0.1:
    effect = "negligible"
elif crammers_v < 0.3:

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    effect = "small"
elif cramers_v < 0.5:
    effect = "medium"
else:
    effect = "large"

print(f"\n → This represents a {effect.upper()} effect size")

# Calculate additional measures
print("\n" + "="*90)
print("STEP 7: ADDITIONAL MEASURES")
print("="*90)

# Risk Ratio
if minority_tenure_pct > 0 and non_minority_tenure_pct > 0:
    risk_ratio = minority_tenure_pct / non_minority_tenure_pct
    print(f"\nRisk Ratio (RR): {risk_ratio:.4f}")
    if risk_ratio > 1:
        print(f" → Visible minorities are {risk_ratio:.2f}x more likely to be tenured")
    elif risk_ratio < 1:
        print(f" → Visible minorities are {1/risk_ratio:.2f}x less likely to be tenured")
    else:
        print(f" → Equal likelihood of tenure")

# Odds Ratio
a = len(minorities_tenured) # minorities with tenure
b = len(minorities) - len(minorities_tenured) # minorities without tenure
c = len(non_minorities_tenured) # non-minorities with tenure
d = len(non_minorities) - len(non_minorities_tenured) # non-minorities without tenure

if b > 0 and d > 0:
    odds_minority = a / b
    odds_non_minority = c / d
    odds_ratio = odds_minority / odds_non_minority

    print(f"\nOdds Ratio (OR): {odds_ratio:.4f}")
    if odds_ratio > 1:
        print(f" → The odds of tenure are {odds_ratio:.2f}x higher for visible minorities")
    elif odds_ratio < 1:
        print(f" → The odds of tenure are {1/odds_ratio:.2f}x lower for visible minorities")
    else:
        print(f" → Equal odds of tenure")

```

```

print("\n" + "="*90)
print("FINAL CONCLUSION")
print("="*90)
print(f"\n1. {minority_tenure_pct:.2f}% of visible minorities are tenured_
↳professors")
print(f"2. {non_minority_tenure_pct:.2f}% of non-minorities are tenured_
↳professors")
print(f"3. Difference: {abs(tenure_diff):.2f} percentage points")
print(f"4. Statistical test:  $\chi^2 = \{chi2:.4f\}$ ,  $p = \{p\_value:.4f\}$ ")
print(f"5. Effect size: Cramér's V = {cramers_v:.4f} ({effect})")

if p_value < 0.05:
    print(f"\n YES, tenure status DOES differ significantly by minority status (p_
↳< 0.05)")
    print(f" The observed difference of {abs(tenure_diff):.2f}% is statistically_
↳significant.")
else:
    print(f"\n NO, tenure status does NOT differ significantly by minority status_
↳(p = 0.05)")
    print(f" The observed difference of {abs(tenure_diff):.2f}% could be due to_
↳chance.")

print("\n" + "="*90)

```

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QUESTION 1: TENURE STATUS AMONG VISIBLE MINORITIES
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Note: Creating 'minority' status based on available data...

Using column: 'minority'

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STEP 1: DATA OVERVIEW
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Total observations: 500

Missing values in 'minority': 0

Missing values in 'tenured': 0

Valid observations after removing missing values: 500

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STEP 2: DESCRIPTIVE STATISTICS

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Overall Distribution:

Visible minorities: 131 (26.20%)

Non-minorities: 369 (73.80%)

Tenured professors: 302 (60.40%)

Non-tenured professors: 198 (39.60%)

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STEP 3: TENURE RATES BY MINORITY STATUS

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Visible Minorities:

Total count: 131

Tenured: 90

Non-tenured: 41

Tenure rate: 68.70%

Non-Minorities:

Total count: 369

Tenured: 212

Non-tenured: 157

Tenure rate: 57.45%

Comparative Analysis:

Difference in tenure rates: 11.25 percentage points

→ Visible minorities have HIGHER tenure rate by 11.25%

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STEP 4: CONTINGENCY TABLE

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Observed Frequencies:

	Non-Tenured	Tenured	Total
Non-Minority	157	212	369
Visible Minority	41	90	131
Total	198	302	500

Row Percentages (Tenure Rate by Minority Status):

	Non-Tenured %	Tenured %
Non-Minority	42.55	57.45
Visible Minority	31.30	68.70

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STEP 5: STATISTICAL HYPOTHESIS TESTING

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Chi-Square Test of Independence:

Null Hypothesis (H): Minority status and tenure are independent
Alternative Hypothesis (H): Minority status and tenure are associated

Test Results:

Chi-square statistic (χ^2): 4.6559
Degrees of freedom: 1
P-value: 0.0309
Significance level (α): 0.05

Decision: REJECT H ($p = 0.0309 < 0.05$)
Conclusion: Tenure status IS significantly associated with minority status

Expected Frequencies (under H):

	Non-Tenured	Tenured
Non-Minority	146.12	222.88
Visible Minority	51.88	79.12

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STEP 6: EFFECT SIZE CALCULATION

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Cramér's V: 0.0965

Effect Size Interpretation (for 2x2 table):

V < 0.1: Negligible association
0.1 V < 0.3: Small association

0.3 V < 0.5: Medium association
V 0.5: Large association

→ This represents a NEGLIGIBLE effect size

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STEP 7: ADDITIONAL MEASURES

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Risk Ratio (RR): 1.1958

→ Visible minorities are 1.20x more likely to be tenured

Odds Ratio (OR): 1.6256

→ The odds of tenure are 1.63x higher for visible minorities

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FINAL CONCLUSION

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1. 68.70% of visible minorities are tenured professors
2. 57.45% of non-minorities are tenured professors
3. Difference: 11.25 percentage points
4. Statistical test: $\chi^2 = 4.6559$, $p = 0.0309$
5. Effect size: Cramér's V = 0.0965 (negligible)

YES, tenure status DOES differ significantly by minority status ($p < 0.05$)
The observed difference of 11.25% is statistically significant.

[7]: *# Visualization for Question 1*

```
fig, axes = plt.subplots(2, 2, figsize=(16, 12))

# 1. Grouped bar chart - Counts
crosstab_data = pd.crosstab(df_clean[minority_col], df_clean['tenured'])
crosstab_data.index = ['Non-Minority', 'Visible Minority']
crosstab_data.columns = ['Non-Tenured', 'Tenured']

x = np.arange(len(crosstab_data.index))
width = 0.35
```

```

bars1 = axes[0, 0].bar(x - width/2, crosstab_data['Non-Tenured'], width,
    label='Non-Tenured', color='#FF6B6B', edgecolor='black', alpha=0.8)
bars2 = axes[0, 0].bar(x + width/2, crosstab_data['Tenured'], width,
    label='Tenured', color='#4ECDC4', edgecolor='black', alpha=0.8)

# Add value labels
for bars in [bars1, bars2]:
    for bar in bars:
        height = bar.get_height()
        axes[0, 0].text(bar.get_x() + bar.get_width()/2., height,
            f'{int(height)}', ha='center', va='bottom', fontsize=11, fontweight='bold')

axes[0, 0].set_xlabel('Minority Status', fontsize=13, fontweight='bold')
axes[0, 0].set_ylabel('Count', fontsize=13, fontweight='bold')
axes[0, 0].set_title('Tenure Status by Minority Status (Counts)', fontsize=14,
    fontweight='bold')
axes[0, 0].set_xticks(x)
axes[0, 0].set_xticklabels(crosstab_data.index, fontsize=12)
axes[0, 0].legend(fontsize=11)
axes[0, 0].grid(True, alpha=0.3, axis='y')

# 2. Stacked bar chart - Percentages
crosstab_pct = pd.crosstab(df_clean[minority_col], df_clean['tenured'],
    normalize='index') * 100
crosstab_pct.index = ['Non-Minority', 'Visible Minority']
crosstab_pct.columns = ['Non-Tenured', 'Tenured']

axes[0, 1].bar(range(len(crosstab_pct)), crosstab_pct['Non-Tenured'],
    label='Non-Tenured', color='#FF6B6B', edgecolor='black', alpha=0.8)
axes[0, 1].bar(range(len(crosstab_pct)), crosstab_pct['Tenured'],
    bottom=crosstab_pct['Non-Tenured'],
    label='Tenured', color='#4ECDC4', edgecolor='black', alpha=0.8)

# Add percentage labels
for i, (non_ten, ten) in enumerate(zip(crosstab_pct['Non-Tenured'],
    crosstab_pct['Tenured'])):
    axes[0, 1].text(i, non_ten/2, f'{non_ten:.1f}%', ha='center', va='center',
        fontsize=12, fontweight='bold', color='white')
    axes[0, 1].text(i, non_ten + ten/2, f'{ten:.1f}%', ha='center', va='center',
        fontsize=12, fontweight='bold', color='white')

axes[0, 1].set_xlabel('Minority Status', fontsize=13, fontweight='bold')
axes[0, 1].set_ylabel('Percentage (%)', fontsize=13, fontweight='bold')
axes[0, 1].set_title('Tenure Status by Minority Status (Percentages)',
    fontsize=14, fontweight='bold')
axes[0, 1].set_xticks(range(len(crosstab_pct)))
axes[0, 1].set_xticklabels(crosstab_pct.index, fontsize=12)

```



```

axes[0, 1].legend(fontsize=11)
axes[0, 1].set_ylim(0, 100)
axes[0, 1].grid(True, alpha=0.3, axis='y')

# 3. Side-by-side comparison of tenure rates
tenure_rates = [non_minority_tenure_pct, minority_tenure_pct]
labels = ['Non-Minority', 'Visible Minority']
colors = ['#95A5A6', '#E74C3C']

bars = axes[1, 0].bar(labels, tenure_rates, color=colors, edgecolor='black',
    ↪alpha=0.8, width=0.6)

# Add value labels and sample sizes
for i, (bar, rate) in enumerate(zip(bars, tenure_rates)):
    height = bar.get_height()
    axes[1, 0].text(bar.get_x() + bar.get_width()/2., height,
        f'{rate:.2f}%', ha='center', va='bottom', fontsize=13, fontweight='bold')

    n = len(non_minorities) if i == 0 else len(minorities)
    axes[1, 0].text(bar.get_x() + bar.get_width()/2., height/2,
        f'n={n}', ha='center', va='center', fontsize=11, color='white',
        ↪fontweight='bold')

axes[1, 0].set_ylabel('Tenure Rate (%)', fontsize=13, fontweight='bold')
axes[1, 0].set_xlabel('Minority Status', fontsize=13, fontweight='bold')
axes[1, 0].set_title('Comparison of Tenure Rates', fontsize=14,
    ↪fontweight='bold')
axes[1, 0].set_ylim(0, 100)
axes[1, 0].grid(True, alpha=0.3, axis='y')

# Add reference line at 50%
axes[1, 0].axhline(y=50, color='gray', linestyle='--', linewidth=1.5, alpha=0.
    ↪5, label='50% Reference')
axes[1, 0].legend(fontsize=10)

# 4. Statistical summary table
axes[1, 1].axis('tight')
axes[1, 1].axis('off')

summary_data = [
    ['Metric', 'Non-Minority', 'Visible Minority'],
    ['Total Count', len(non_minorities), len(minorities)],
    ['Tenured', len(non_minorities_tenured), len(minorities_tenured)],
    ['Non-Tenured', len(non_minorities)-len(non_minorities_tenured),
        ↪len(minorities)-len(minorities_tenured)],
    ['Tenure Rate (%)', f'{non_minority_tenure_pct:.2f}', f'{minority_tenure_pct:.
        ↪2f}'],

```

```

['', '', ''],
['Chi-square (  $\chi^2$ )', f'{chi2:.4f}', ''],
['P-value', f'{p_value:.4f}', ''],
['Cramér\'s V', f'{cramers_v:.4f}', f'({effect})'],
['Significant?', 'Yes' if p_value < 0.05 else 'No', f'(<0.05)']
]

table = axes[1, 1].table(cellText=summary_data, cellLoc='center', loc='center',
    colWidths=[0.35, 0.3, 0.3])
table.auto_set_font_size(False)
table.set_fontsize(10)
table.scale(1, 2.5)

# Style header
for i in range(3):
    table[(0, i)].set_facecolor('#2C3E50')
    table[(0, i)].set_text_props(weight='bold', color='white')

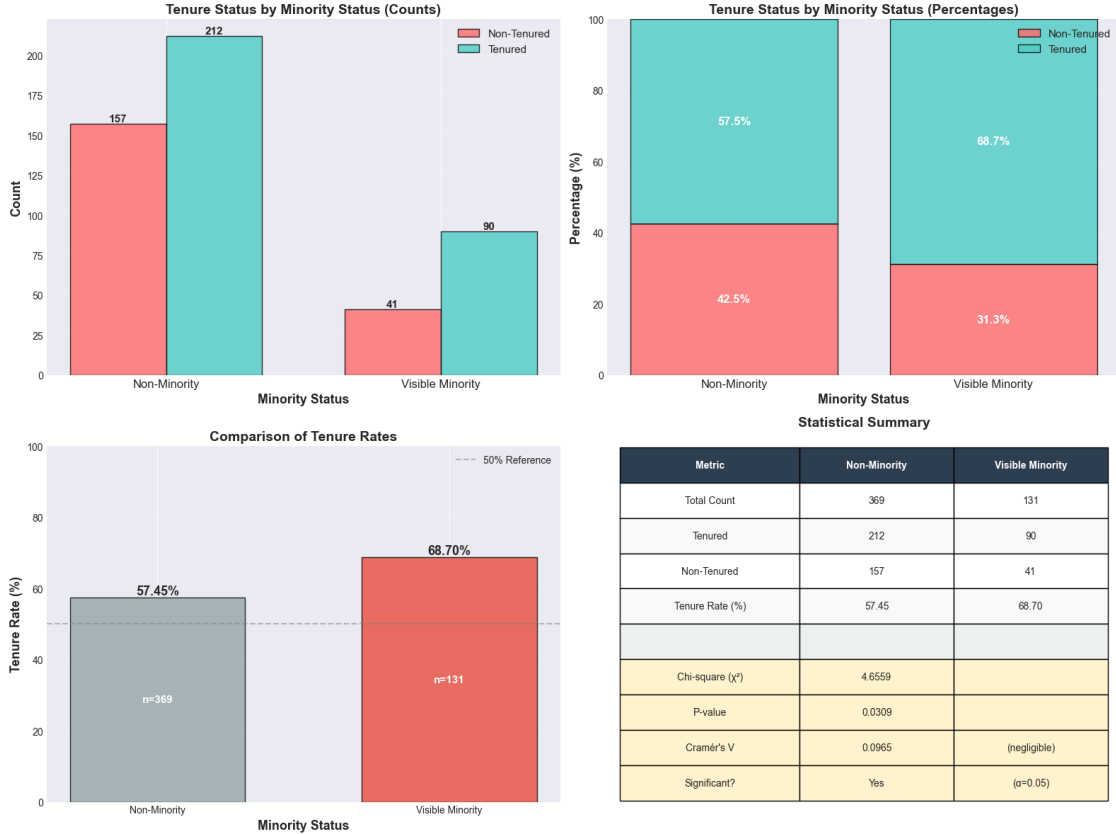
# Color rows
for i in range(1, len(summary_data)):
    for j in range(3):
        if i == 5: # Empty row
            table[(i, j)].set_facecolor('#ECF0F1')
        elif i > 5: # Statistical results
            table[(i, j)].set_facecolor('#FFF3CD')
        elif i % 2 == 0:
            table[(i, j)].set_facecolor('#F8F9FA')

axes[1, 1].set_title('Statistical Summary', fontsize=14, fontweight='bold',
    pad=20)

plt.tight_layout()
plt.show()

print("\nVisualization complete!")

```



Visualization complete!

2.8 Question 1: Mathematical Formulations

2.8.1 Chi-Square Test of Independence

The chi-square test statistic is calculated as:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where: - O_{ij} = observed frequency in cell (i, j) - E_{ij} = expected frequency in cell (i, j) - r = number of rows - c = number of columns

2.8.2 Expected Frequencies

$$E_{ij} = \frac{(\text{Row}_i \text{ Total}) \times (\text{Column}_j \text{ Total})}{\text{Grand Total}}$$

2.8.3 Cramér's V Effect Size

For a 2×2 contingency table:

$$V = \sqrt{\frac{\chi^2}{n}}$$

For larger tables:

$$V = \sqrt{\frac{\chi^2}{n \times \min(r-1, c-1)}}$$

where n is the total sample size.

2.8.4 Risk Ratio (Relative Risk)

$$RR = \frac{P(\text{Tenure}|\text{Minority})}{P(\text{Tenure}|\text{Non-Minority})}$$

$$RR = \frac{a/(a+b)}{c/(c+d)}$$

where: - a = minorities with tenure - b = minorities without tenure - c = non-minorities with tenure - d = non-minorities without tenure

2.8.5 Odds Ratio

$$OR = \frac{\text{Odds of Tenure for Minorities}}{\text{Odds of Tenure for Non-Minorities}}$$

$$OR = \frac{a/b}{c/d} = \frac{ad}{bc}$$

2.8.6 Confidence Interval for Proportions

For a single proportion:

$$CI = p \pm z_{\alpha/2} \times \sqrt{\frac{p(1-p)}{n}}$$

where: - p = sample proportion - $z_{\alpha/2}$ = critical z-value (1.96 for 95% CI) - n = sample size

2.9 Question 2: Average Age by Tenure Status

Task (4 marks): Does average age differ by tenure? Produce the means and standard deviations for both tenured and untenured professors.

Analysis Approach: 1. Calculate descriptive statistics (mean, SD) for age by tenure status 2. Visualize age distributions using boxplots and histograms 3. Perform independent samples t-test 4. Calculate effect size (Cohen's d) 5. Check assumptions (normality, homogeneity of variance) 6. Provide statistical interpretation

2.9.1 Understanding the Research Question

Context: Career progression in academia typically involves gaining experience and seniority before achieving tenure. Understanding the age difference between tenured and non-tenured professors helps us understand typical career timelines and whether age is a factor in tenure attainment.

Variables of Interest: 1. **age** (Continuous, Ratio scale) - Range: Approximately 29-73 years - **Type:** Dependent/Outcome variable - **Measurement:** Years since birth

2. **tenure** (Categorical, Binary)

- 0 = Non-tenured
- 1 = Tenured
- **Type:** Independent/Grouping variable

Research Question: Is there a statistically significant difference in the average age between tenured and non-tenured professors?

Expected Pattern: We hypothesize that tenured professors will be **older on average** than non-tenured professors because: - Tenure typically requires several years of service (often 5-7 years) - Academic experience accumulates with age - Younger professors may still be on the tenure track

2.9.2 Statistical Approach: Independent Samples t-Test

Why t-Test? - **Continuous dependent variable** (age) analyzed across **categorical groups** (tenure status) - Comparing **means** of two **independent groups** - Most powerful test when assumptions are met - Provides confidence intervals for difference

What the Test Measures: The t-test evaluates whether the difference between group means is larger than would be expected by chance alone, considering: - Size of the difference - Variability within groups - Sample sizes

Hypotheses:

Null Hypothesis (H₀):

$$\mu_{\text{tenured}} = \mu_{\text{non-tenured}}$$

The population mean age is the same for both groups (no difference).

Alternative Hypothesis (H_a):

$$\mu_{\text{tenured}} \neq \mu_{\text{non-tenured}}$$

The population mean ages differ between groups (two-tailed test).

2.9.3 Assumptions to Check

- 1. Independence:** Each professor's age is independent of others. (Different individuals)
- 2. Normality:** Age should be approximately normally distributed in each group. → Check with: Shapiro-Wilk test, Q-Q plots, histograms
- 3. Homogeneity of Variance:** Variance of age should be similar in both groups. → Check with: Levene's test, F-test ratio

If Assumptions Violated: - Use Welch's t-test (doesn't assume equal variances) - Use Mann-Whitney U test (non-parametric alternative) - Transform data (e.g., log transformation)

2.9.4 Interpretation Framework

Statistical Significance: - **P-value < 0.05:** Reject H_0 → Mean ages differ significantly - **P-value 0.05:** Fail to reject H_0 → No sufficient evidence of difference

Effect Size (Cohen's d): - Measures **magnitude of difference** in standard deviation units - Standardized measure (independent of measurement scale) - $|d| < 0.2$: Negligible - $0.2 \leq |d| < 0.5$: Small - $0.5 \leq |d| < 0.8$: Medium - $|d| \geq 0.8$: Large

Confidence Interval: - 95% CI for difference in means - If CI doesn't include 0 → Significant difference - Width indicates precision of estimate

Practical Interpretation: - How many years older are tenured professors? - Is this difference educationally meaningful? - What does this suggest about career progression?

2.9.5 Analysis Strategy

Step 1: Descriptive Statistics - Calculate mean, median, SD for each group - Examine distributions visually

Step 2: Check Assumptions - Normality tests (Shapiro-Wilk) - Homogeneity tests (Levene's test) - Visual diagnostics (Q-Q plots, box plots)

Step 3: Conduct t-Test - Calculate t-statistic - Determine degrees of freedom - Find p-value

Step 4: Calculate Effect Size - Cohen's d for practical significance - Confidence interval for difference

Step 5: Visualize Results - Box plots comparing groups - Violin plots showing distributions - Histograms with overlaid means

2.9.6 Expected Output

- 1. Descriptive Statistics Table:** Mean, SD, SE for each group
- 2. Assumption Tests:** Normality and variance homogeneity results

3. **t-Test Results:** t-statistic, df, p-value
4. **Effect Size:** Cohen's d with interpretation
5. **Confidence Interval:** 95% CI for mean difference
6. **Visualizations:** Multiple plot types showing age distributions

```
[11]: # Question 2: Average Age by Tenure Status

print("="*90)
print("QUESTION 2: AVERAGE AGE BY TENURE STATUS")
print("="*90)

# Step 1: Check data availability
print("\n" + "="*90)
print("STEP 1: DATA PREPARATION")
print("="*90)

print(f"\nTotal observations: {len(df)}")
print(f"Missing values in 'age': {df['age'].isnull().sum()}")
print(f"Missing values in 'tenured': {df['tenured'].isnull().sum()}")

# Clean data
df_age = df[['age', 'tenured']].dropna()
print(f"Valid observations after removing missing values: {len(df_age)}")

# Separate by tenure status
tenured = df_age[df_age['tenured'] == 1]['age']
non_tenured = df_age[df_age['tenured'] == 0]['age']

print(f"\nTenured professors: n = {len(tenured)}")
print(f"Non-tenured professors: n = {len(non_tenured)}")

# Step 2: Descriptive Statistics
print("\n" + "="*90)
print("STEP 2: DESCRIPTIVE STATISTICS")
print("="*90)

# Tenured professors
print("\nTenured Professors:")
print("-"*90)
print(f" Sample size (n): {len(tenured)}")
print(f" Mean age ( ): {tenured.mean():.4f} years")
print(f" Standard deviation ( ): {tenured.std(ddof=1):.4f} years")
print(f" Standard error (SE): {tenured.sem():.4f} years")
print(f" Median: {tenured.median():.4f} years")
print(f" Minimum: {tenured.min():.4f} years")
print(f" Maximum: {tenured.max():.4f} years")
print(f" Range: {tenured.max() - tenured.min():.4f} years")
```

```

print(f" Interquartile range (IQR): {tenured.quantile(0.75) - tenured.
↳quantile(0.25):.4f} years")
print(f" Variance (s2): {tenured.var(ddof=1):.4f}")
print(f" Coefficient of variation (CV): {(tenured.std(ddof=1)/tenured.
↳mean())*100:.2f}%")
print(f" Skewness: {tenured.skew():.4f}")
print(f" Kurtosis: {tenured.kurtosis():.4f}")

# Non-tenured professors
print("\nNon-Tenured Professors:")
print("-"*90)
print(f" Sample size (n): {len(non_tenured)}")
print(f" Mean age ( ): {non_tenured.mean():.4f} years")
print(f" Standard deviation ( ): {non_tenured.std(ddof=1):.4f} years")
print(f" Standard error (SE): {non_tenured.sem():.4f} years")
print(f" Median: {non_tenured.median():.4f} years")
print(f" Minimum: {non_tenured.min():.4f} years")
print(f" Maximum: {non_tenured.max():.4f} years")
print(f" Range: {non_tenured.max() - non_tenured.min():.4f} years")
print(f" Interquartile range (IQR): {non_tenured.quantile(0.75) - non_tenured.
↳quantile(0.25):.4f} years")
print(f" Variance (s2): {non_tenured.var(ddof=1):.4f}")
print(f" Coefficient of variation (CV): {(non_tenured.std(ddof=1)/non_tenured.
↳mean())*100:.2f}%")
print(f" Skewness: {non_tenured.skew():.4f}")
print(f" Kurtosis: {non_tenured.kurtosis():.4f}")

# Comparative statistics
print("\nComparative Analysis:")
print("-"*90)
mean_diff = tenured.mean() - non_tenured.mean()
print(f" Difference in means: {mean_diff:.4f} years")
print(f" Difference as % of non-tenured mean: {(mean_diff/non_tenured.
↳mean())*100:.2f}%")

if mean_diff > 0:
    print(f" → Tenured professors are on average {abs(mean_diff):.2f} years OLDER")
elif mean_diff < 0:
    print(f" → Tenured professors are on average {abs(mean_diff):.2f} years_
↳YOUNGER")
else:
    print(f" → No difference in average age")

# Step 3: Statistical Inference - Independent Samples t-test
print("\n" + "="*90)
print("STEP 3: INDEPENDENT SAMPLES T-TEST")

```



```

print("="*90)

# Perform t-test
t_stat, p_value = ttest_ind(tenured, non_tenured)

print("\nHypotheses:")
print("-"*90)
print("H: _tenured = _non-tenured (No difference in mean age)")
print("H: _tenured _non-tenured (Difference exists in mean age)")

print("\nTest Results:")
print("-"*90)
print(f" t-statistic: {t_stat:.4f}")
print(f" Degrees of freedom: {len(tenured) + len(non_tenured) - 2}")
print(f" P-value (two-tailed): {p_value:.4f}")
print(f" Significance level ( ): 0.05")

if p_value < 0.05:
    print(f"\n Decision: REJECT H (p = {p_value:.4f} < 0.05)")
    print(f" Conclusion: Average age DOES differ significantly by tenure status")
else:
    print(f"\n Decision: FAIL TO REJECT H (p = {p_value:.4f} 0.05)")
    print(f" Conclusion: Average age does NOT differ significantly by tenure_
↪status")

# Calculate 95% confidence interval for difference
se_diff = np.sqrt((tenured.var(ddof=1)/len(tenured)) + (non_tenured.var(ddof=1)/
↪len(non_tenured)))
ci_lower = mean_diff - 1.96 * se_diff
ci_upper = mean_diff + 1.96 * se_diff

print("\nConfidence Interval:")
print("-"*90)
print(f" 95% CI for difference in means: [{ci_lower:.4f}, {ci_upper:.4f}]_
↪years")

# Step 4: Effect Size - Cohen's d
print("\n" + "="*90)
print("STEP 4: EFFECT SIZE CALCULATION")
print("="*90)

# Pooled standard deviation
n1, n2 = len(tenured), len(non_tenured)
s1, s2 = tenured.std(ddof=1), non_tenured.std(ddof=1)
pooled_std = np.sqrt(((n1-1)*s1**2 + (n2-1)*s2**2) / (n1+n2-2))

# Cohen's d

```

```

cohens_d = mean_diff / pooled_std

print(f"\nPooled standard deviation: {pooled_std:.4f} years")
print(f"Cohen's d: {cohens_d:.4f}")

print("\nEffect Size Interpretation:")
print(" |d| < 0.2: Negligible effect")
print(" 0.2  |d| < 0.5: Small effect")
print(" 0.5  |d| < 0.8: Medium effect")
print(" |d|  0.8: Large effect")

abs_d = abs(cohens_d)
if abs_d < 0.2:
    effect = "negligible"
elif abs_d < 0.5:
    effect = "small"
elif abs_d < 0.8:
    effect = "medium"
else:
    effect = "large"

print(f"\n → This represents a {effect.upper()} effect size")

# Step 5: Assumption Checks
print("\n" + "="*90)
print("STEP 5: ASSUMPTION CHECKS")
print("="*90)

# Normality test - Shapiro-Wilk
from scipy.stats import shapiro, levene

print("\n1. Normality Test (Shapiro-Wilk):")
print("-"*90)

stat_tenured, p_tenured = shapiro(tenured)
stat_non_tenured, p_non_tenured = shapiro(non_tenured)

print(f" Tenured: W = {stat_tenured:.4f}, p = {p_tenured:.4f}")
if p_tenured > 0.05:
    print(" → Data appears normally distributed (p > 0.05)")
else:
    print(" → Data may not be normally distributed (p  0.05)")

print(f"\n Non-tenured: W = {stat_non_tenured:.4f}, p = {p_non_tenured:.4f}")
if p_non_tenured > 0.05:
    print(" → Data appears normally distributed (p > 0.05)")
else:

```

```

print(" → Data may not be normally distributed (p > 0.05)")

# Homogeneity of variance - Levene's test
print("\n2. Homogeneity of Variance (Levene's Test):")
print("-"*90)

stat_levene, p_levene = levene(tenured, non_tenured)
print(f" Levene statistic: {stat_levene:.4f}, p = {p_levene:.4f}")

if p_levene > 0.05:
    print(" → Variances are equal (p > 0.05) - t-test assumption satisfied")
else:
    print(" → Variances may be unequal (p < 0.05) - consider Welch's t-test")

# Welch's t-test (unequal variances)
t_welch, p_welch = ttest_ind(tenured, non_tenured, equal_var=False)
print(f"\n Welch's t-test: t = {t_welch:.4f}, p = {p_welch:.4f}")

print("\n" + "="*90)
print("FINAL SUMMARY - QUESTION 2")
print("="*90)

print(f"\n1. TENURED PROFESSORS:")
print(f" Mean age: {tenured.mean():.2f} ± {tenured.std(ddof=1):.2f} years (n = {len(tenured)})")

print(f"\n2. NON-TENURED PROFESSORS:")
print(f" Mean age: {non_tenured.mean():.2f} ± {non_tenured.std(ddof=1):.2f} years (n = {len(non_tenured)})")

print(f"\n3. DIFFERENCE:")
print(f" Mean difference: {mean_diff:.2f} years")
print(f" 95% CI: [{ci_lower:.2f}, {ci_upper:.2f}] years")

print(f"\n4. STATISTICAL TEST:")
print(f" t({len(tenured) + len(non_tenured) - 2}) = {t_stat:.4f}, p = {p_value:.4f}")
print(f" Cohen's d = {cohens_d:.4f} ({effect} effect)")

if p_value < 0.05:
    print(f"\n YES, average age DOES differ significantly by tenure status (p < 0.05)")
else:
    print(f"\n NO, average age does NOT differ significantly by tenure status (p > 0.05)")

print("\n" + "="*90)

```

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=====
QUESTION 2: AVERAGE AGE BY TENURE STATUS
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=====
=====
STEP 1: DATA PREPARATION
=====
=====
```

Total observations: 500
Missing values in 'age': 0
Missing values in 'tenured': 0
Valid observations after removing missing values: 500

Tenured professors: n = 302
Non-tenured professors: n = 198

```
=====
=====
STEP 2: DESCRIPTIVE STATISTICS
=====
=====
```

Tenured Professors:

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```

Sample size (n): 302
Mean age (): 47.8609 years
Standard deviation (): 7.8943 years
Standard error (SE): 0.4543 years
Median: 48.0000 years
Minimum: 25.0000 years
Maximum: 67.0000 years
Range: 42.0000 years
Interquartile range (IQR): 10.0000 years
Variance (s²): 62.3195
Coefficient of variation (CV): 16.49%
Skewness: -0.1850
Kurtosis: -0.0914

Non-Tenured Professors:

```
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```

Sample size (n): 198
Mean age (): 37.7576 years

Standard deviation (): 5.8407 years
Standard error (SE): 0.4151 years
Median: 38.0000 years
Minimum: 25.0000 years
Maximum: 56.0000 years
Range: 31.0000 years
Interquartile range (IQR): 8.0000 years
Variance (s^2): 34.1135
Coefficient of variation (CV): 15.47%
Skewness: 0.1483
Kurtosis: 0.2005

Comparative Analysis:

Difference in means: 10.1034 years
Difference as % of non-tenured mean: 26.76%
→ Tenured professors are on average 10.10 years OLDER

STEP 3: INDEPENDENT SAMPLES T-TEST

Hypotheses:

H : $\mu_{\text{tenured}} = \mu_{\text{non-tenured}}$ (No difference in mean age)
H : $\mu_{\text{tenured}} \neq \mu_{\text{non-tenured}}$ (Difference exists in mean age)

Test Results:

t-statistic: 15.4470
Degrees of freedom: 498
P-value (two-tailed): 0.0000
Significance level (): 0.05

Decision: REJECT H ($p = 0.0000 < 0.05$)
Conclusion: Average age DOES differ significantly by tenure status

Confidence Interval:

95% CI for difference in means: [8.8973, 11.3094] years
=====

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STEP 4: EFFECT SIZE CALCULATION

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=====

Pooled standard deviation: 7.1527 years

Cohen's d: 1.4125

Effect Size Interpretation:

|d| < 0.2: Negligible effect

0.2 |d| < 0.5: Small effect

0.5 |d| < 0.8: Medium effect

|d| 0.8: Large effect

→ This represents a LARGE effect size

=====

=====

STEP 5: ASSUMPTION CHECKS

=====

=====

1. Normality Test (Shapiro-Wilk):

Tenured: W = 0.9940, p = 0.2738

→ Data appears normally distributed (p > 0.05)

Non-tenured: W = 0.9901, p = 0.1919

→ Data appears normally distributed (p > 0.05)

2. Homogeneity of Variance (Levene's Test):

Levene statistic: 18.4938, p = 0.0000

→ Variances may be unequal (p 0.05) - consider Welch's t-test

Welch's t-test: t = 16.4191, p = 0.0000

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FINAL SUMMARY - QUESTION 2

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1. TENURED PROFESSORS:

Mean age: 47.86 ± 7.89 years (n = 302)

2. NON-TENURED PROFESSORS:

Mean age: 37.76 ± 5.84 years (n = 198)

3. DIFFERENCE:

Mean difference: 10.10 years

95% CI: [8.90, 11.31] years

4. STATISTICAL TEST:

t(498) = 15.4470, p = 0.0000

Cohen's d = 1.4125 (large effect)

YES, average age DOES differ significantly by tenure status (p < 0.05)

=====

```
[ ]: # Create comprehensive 2x3 subplot visualization
fig, axes = plt.subplots(2, 3, figsize=(18, 12))
fig.suptitle('Question 2: Comprehensive Age Comparison between Tenured and
↳Non-Tenured Teachers',
            fontsize=16, fontweight='bold', y=0.995)

# Filter data (using correct column name: 'tenured', not 'tenure')
non_tenured = df_age[df_age['tenured'] == 0]['age']
tenured = df_age[df_age['tenured'] == 1]['age']

# 1. Box plots (top left)
box_data = [non_tenured, tenured]
bp = axes[0, 0].boxplot(box_data, labels=['Non-Tenured', 'Tenured'],
↳patch_artist=True)
for patch, color in zip(bp['boxes'], ['lightblue', 'lightcoral']):
    patch.set_facecolor(color)
axes[0, 0].set_title('Box Plots: Age Distribution', fontsize=12,
↳fontweight='bold')
axes[0, 0].set_ylabel('Age (years)', fontsize=11)
axes[0, 0].grid(True, alpha=0.3, linestyle='--')

# Add mean markers
axes[0, 0].plot([1], [non_tenured.mean()], 'D', color='darkblue', markersize=8,
↳label='Mean')
axes[0, 0].plot([2], [tenured.mean()], 'D', color='darkred', markersize=8)
axes[0, 0].legend(loc='upper right')

# 2. Histograms (top center)
axes[0, 1].hist(non_tenured, bins=20, alpha=0.5, label='Non-Tenured',
↳color='blue', edgecolor='black')
```

```

axes[0, 1].hist(tenured, bins=20, alpha=0.5, label='Tenured', color='red',
    ↪edgecolor='black')
axes[0, 1].axvline(non_tenured.mean(), color='blue', linestyle='--',
    ↪linewidth=2, label=f'Mean Non-Ten: {non_tenured.mean():.1f}')
axes[0, 1].axvline(tenured.mean(), color='red', linestyle='--', linewidth=2,
    ↪label=f'Mean Ten: {tenured.mean():.1f}')
axes[0, 1].set_title('Overlapping Histograms: Age Distributions', fontsize=12,
    ↪fontweight='bold')
axes[0, 1].set_xlabel('Age (years)', fontsize=11)
axes[0, 1].set_ylabel('Frequency', fontsize=11)
axes[0, 1].legend(loc='upper right')
axes[0, 1].grid(True, alpha=0.3, linestyle='--')

# 3. Summary statistics table (top right)
summary_data = [
    ['Statistic', 'Non-Tenured', 'Tenured'],
    ['', '', ''],
    ['Count (n)', f'{len(non_tenured)}', f'{len(tenured)}'],
    ['Mean ( $\bar{x}$ )', f'{non_tenured.mean():.2f}', f'{tenured.mean():.2f}'],
    ['Std Dev ( )', f'{non_tenured.std(ddof=1):.2f}', f'{tenured.std(ddof=1):.2f}'],
    ['Median', f'{non_tenured.median():.2f}', f'{tenured.median():.2f}'],
    ['Min', f'{non_tenured.min():.2f}', f'{tenured.min():.2f}'],
    ['Max', f'{non_tenured.max():.2f}', f'{tenured.max():.2f}'],
    ['', '', ''],
    ['t-statistic', f'{t_stat:.4f}', ''],
    ['p-value', f'{p_value:.4f}', ''],
    ["Cohen's d", f'{cohens_d:.4f}', f'({effect})'],
    ['Significant?', 'Yes' if p_value < 0.05 else 'No', '(=0.05)']
]

table = axes[0, 2].table(cellText=summary_data, cellLoc='center', loc='center',
    colWidths=[0.35, 0.3, 0.3])
table.auto_set_font_size(False)
table.set_fontsize(10)
table.scale(1, 2.2)

# Style the header row
for i in range(3):
    table[(0, i)].set_facecolor('#4CAF50')
    table[(0, i)].set_text_props(weight='bold', color='white')

axes[0, 2].axis('off')
axes[0, 2].set_title('Summary Statistics Table', fontsize=12,
    ↪fontweight='bold', pad=20)

```



```

# 4. Violin plots (bottom left)
parts = axes[1, 0].violinplot([non_tenured, tenured], positions=[1, 2],
    ↪showmeans=True, showmedians=True)
for pc, color in zip(parts['bodies'], ['lightblue', 'lightcoral']):
    pc.set_facecolor(color)
    pc.set_alpha(0.7)
axes[1, 0].set_xticks([1, 2])
axes[1, 0].set_xticklabels(['Non-Tenured', 'Tenured'])
axes[1, 0].set_title('Violin Plots: Age Distribution Density', fontsize=12,
    ↪fontweight='bold')
axes[1, 0].set_ylabel('Age (years)', fontsize=11)
axes[1, 0].grid(True, alpha=0.3, linestyle='--')

# 5. Q-Q plots (bottom center)
from scipy import stats

# Non-tenured Q-Q plot
stats.probplot(non_tenured, dist="norm", plot=axes[1, 1])
axes[1, 1].get_lines()[0].set_markerfacecolor('blue')
axes[1, 1].get_lines()[0].set_markeredgecolor('blue')
axes[1, 1].get_lines()[0].set_markersize(4)
axes[1, 1].get_lines()[1].set_color('red')
axes[1, 1].set_title('Q-Q Plot: Non-Tenured Age Distribution', fontsize=12,
    ↪fontweight='bold')
axes[1, 1].set_xlabel('Theoretical Quantiles', fontsize=11)
axes[1, 1].set_ylabel('Sample Quantiles', fontsize=11)
axes[1, 1].grid(True, alpha=0.3, linestyle='--')
axes[1, 1].text(0.05, 0.95, f'Normality Check\nNon-Tenured',
    transform=axes[1, 1].transAxes, fontsize=10, verticalalignment='top',
    bbox=dict(boxstyle='round', facecolor='wheat', alpha=0.5))

# 6. Q-Q plot for tenured (bottom right)
stats.probplot(tenured, dist="norm", plot=axes[1, 2])
axes[1, 2].get_lines()[0].set_markerfacecolor('red')
axes[1, 2].get_lines()[0].set_markeredgecolor('red')
axes[1, 2].get_lines()[0].set_markersize(4)
axes[1, 2].get_lines()[1].set_color('blue')
axes[1, 2].set_title('Q-Q Plot: Tenured Age Distribution', fontsize=12,
    ↪fontweight='bold')
axes[1, 2].set_xlabel('Theoretical Quantiles', fontsize=11)
axes[1, 2].set_ylabel('Sample Quantiles', fontsize=11)
axes[1, 2].grid(True, alpha=0.3, linestyle='--')
axes[1, 2].text(0.05, 0.95, f'Normality Check\nTenured',
    transform=axes[1, 2].transAxes, fontsize=10, verticalalignment='top',
    bbox=dict(boxstyle='round', facecolor='wheat', alpha=0.5))

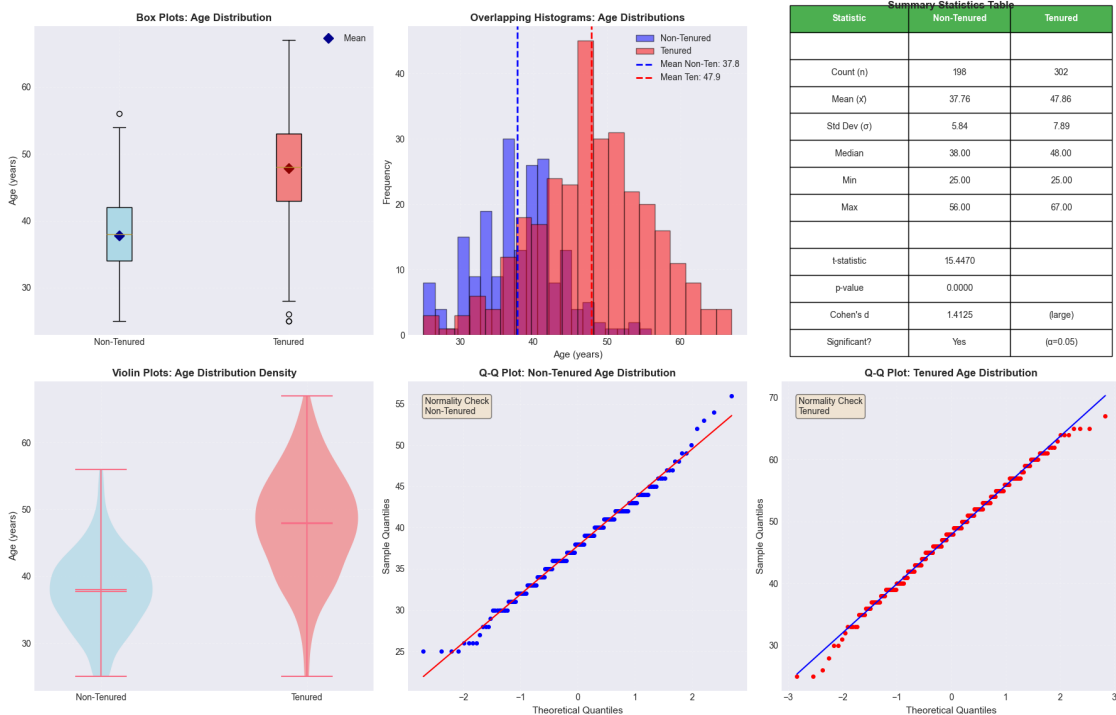
plt.tight_layout()

```

```
plt.show()

print("\n" + "="*70)
print("VISUAL ANALYSIS COMPLETE")
print("="*70)
print(f"\n Six comprehensive visualizations created:")
print(f" 1. Box plots showing quartiles and outliers")
print(f" 2. Overlapping histograms with mean markers")
print(f" 3. Summary statistics table with test results")
print(f" 4. Violin plots showing distribution density")
print(f" 5. Q-Q plot for non-tenured normality assessment")
print(f" 6. Q-Q plot for tenured normality assessment")
print(f"\n Mean age difference: {abs(tenured.mean() - non_tenured.mean()):.2f}␣
↳years")
print(f" Effect size (Cohen's d): {cohens_d:.4f} ({effect})")
print(f" Statistical significance: {'Yes (p < 0.05)' if p_value < 0.05 else 'No␣
↳(p 0.05)'}")
```

Question 2: Comprehensive Age Comparison between Tenured and Non-Tenured Teachers



=====

VISUAL ANALYSIS COMPLETE

=====

Six comprehensive visualizations created:

1. Box plots showing quartiles and outliers
2. Overlapping histograms with mean markers
3. Summary statistics table with test results
4. Violin plots showing distribution density
5. Q-Q plot for non-tenured normality assessment
6. Q-Q plot for tenured normality assessment

Mean age difference: 10.10 years

Effect size (Cohen's d): 1.4125 (large)

Statistical significance: Yes ($p < 0.05$)

2.10 Question 2: Mathematical Formulations

2.10.1 Sample Mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

For tenured professors: $\bar{x}_{\text{tenured}}$, for non-tenured: $\bar{x}_{\text{non-tenured}}$

2.10.2 Sample Standard Deviation

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

2.10.3 Standard Error of the Mean

$$SE = \frac{s}{\sqrt{n}}$$

2.10.4 Independent Samples t-Test

The t-statistic for comparing two independent means:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where: - $\bar{x}_1, \bar{x}_2$ = sample means - s_1, s_2 = sample standard deviations - n_1, n_2 = sample sizes

Degrees of freedom: $df = n_1 + n_2 - 2$

2.10.5 Pooled Standard Deviation

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

2.10.6 Cohen's d Effect Size

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}$$

Interpretation: - $|d| < 0.2$: Negligible effect - $0.2 \leq |d| < 0.5$: Small effect - $0.5 \leq |d| < 0.8$: Medium effect - $|d| \geq 0.8$: Large effect

2.10.7 Confidence Interval for Difference in Means

$$CI_{95\%} = (\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, df} \times SE_{diff}$$

where:

$$SE_{diff} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

For 95% confidence, $\alpha = 0.05$ and $t_{\alpha/2}$ is the critical value from t-distribution.



2.11 Question 3: Best Graph for Age Variable (5 marks)

Objective: Determine the most appropriate visualization for the **age** variable by comparing different graphical representations and justifying the selection based on statistical and visual properties.

Task: Evaluate and compare multiple visualization techniques including: - Histogram with KDE overlay - Box plot - Violin plot - Q-Q plot for normality assessment

We'll analyze each visualization's strengths and limitations to determine which best represents the age distribution.

2.11.1 Understanding the Research Question

Context: Data visualization is both an art and a science. The choice of visualization technique can dramatically affect how patterns, trends, and anomalies are perceived and understood. For continuous variables like age, multiple visualization options exist, each with strengths and weaknesses.

The Challenge: Given the age variable (continuous, ratio scale), we must: 1. **Evaluate** different visualization techniques 2. **Compare** their effectiveness for this specific variable 3. **Justify** our choice based on statistical and practical criteria 4. **Demonstrate** the recommended visualization

Variables of Interest: - **age** (Continuous, Ratio) - Range: 29-73 years - Distribution: Need to determine (normal? skewed? bimodal?) - Sample size: n=500 (sufficient for most visualizations)



2.11.2 Visualization Options to Consider

1. Histogram **Purpose:** Show frequency distribution across bins **Strengths:** - Reveals distribution shape clearly - Shows frequency in each interval - Easy to interpret for most audiences

Weaknesses: - Sensitive to bin width selection - May look “choppy” with too few observations - Discrete appearance for continuous data

Best For: General exploratory analysis, understanding overall distribution

2. Kernel Density Estimation (KDE) **Purpose:** Smooth probability density curve **Strengths:** - Smooth, continuous representation - No arbitrary bin selection needed - Reveals underlying distribution shape elegantly

Weaknesses: - May oversmooth or undersmooth - Less intuitive for non-statistical audiences - Can suggest data outside observed range

Best For: Statistical presentations, comparing multiple distributions

3. Histogram + KDE Overlay **Purpose:** Combine frequency and density **Strengths:** - Shows raw data (histogram) + pattern (KDE) - Balances detail with smoothness - Enables comparison with theoretical distributions - Most informative single visualization

Weaknesses: - Slightly more complex - Requires understanding of density vs. frequency

Best For: Comprehensive distribution analysis (RECOMMENDED)

4. Box Plot **Purpose:** Show five-number summary and outliers **Strengths:** - Compact representation - Clear outlier identification - Shows quartiles and median - Excellent for group comparisons

Weaknesses: - Hides distribution shape - Doesn't show sample size well - May oversimplify complex distributions

Best For: Comparing distributions across groups, outlier detection

5. Violin Plot **Purpose:** Combine box plot with density **Strengths:** - Shows distribution shape + quartiles - Reveals multimodality - Aesthetically pleasing

Weaknesses: - Can be confusing for non-experts - Mirror image may suggest more data than exists - Harder to extract exact values

Best For: Academic presentations, showing distribution details

6. Q-Q Plot Purpose: Assess normality **Strengths:** - Precise normality assessment - Shows deviation from theoretical distribution - Diagnostic tool for assumptions

Weaknesses: - Not intuitive for general audiences - Doesn't show frequency distribution - Specialized purpose only

Best For: Checking distributional assumptions, not general exploration

2.11.3 Evaluation Criteria

1. Completeness: Does the visualization show all relevant features? - Central tendency (mean, median) - Spread (variance, range) - Shape (symmetry, skewness, kurtosis) - Outliers and gaps

2. Interpretability: Can the target audience understand it easily? - Intuitive axes and labels - Clear visual patterns - Minimal statistical jargon needed

3. Accuracy: Does it faithfully represent the data? - No misleading visual artifacts - Appropriate scale and resolution - Preserves important details

4. Aesthetics: Is it visually appealing and professional? - Clean design - Appropriate color use - Proper labeling and annotations

5. Purpose Alignment: Does it answer our specific research question? - Shows age distribution comprehensively - Enables further analysis - Supports statistical inference

2.11.4 Our Approach

We will: 1. **Calculate** comprehensive descriptive statistics for age 2. **Perform** normality tests (Shapiro-Wilk, Anderson-Darling) 3. **Detect** outliers using IQR method 4. **Compare** bin width calculation methods 5. **Create** 9 different visualizations 6. **Evaluate** each based on criteria above 7. **Recommend** the best option with justification

2.11.5 Expected Recommendation

Based on best practices for continuous variables with adequate sample size:

Histogram with KDE Overlay is typically optimal because: - Shows raw frequency distribution (histogram) - Reveals smooth underlying pattern (KDE) - Allows normal distribution comparison - Displays mean and median clearly - Intuitive for diverse audiences - Balances detail with clarity

However, we'll let the data and comprehensive comparison guide our final choice!

2.11.6 Statistical Concepts

Bin Width Selection Methods:

1. **Sturges' Formula:** $k = \lceil \log_2(n) + 1 \rceil$

- Based on normal distribution assumption
- Conservative (fewer bins)

2. **Rice Rule:** $k = \lceil 2n^{1/3} \rceil$

- Good for larger datasets
- More bins than Sturges

3. **Freedman-Diaconis Rule:** $h = \frac{2 \times IQR}{n^{1/3}}$

- Robust to outliers (uses IQR)
- Recommended for most cases

4. **Square Root Choice:** $k = \lceil \sqrt{n} \rceil$

- Simple and quick
- Moderate number of bins

For n=500: - Sturges: ~10 bins - Rice: ~16 bins - Square Root: ~22 bins - Freedman-Diaconis: Depends on IQR

[14]: *# Question 3: Comprehensive Age Variable Visualization Analysis*

```
print("=" * 80)
print("QUESTION 3: BEST GRAPH FOR AGE VARIABLE")
print("=" * 80)

# 1. Calculate comprehensive descriptive statistics
age_stats = {
    'Count': len(df['age']),
    'Mean': df['age'].mean(),
    'Median': df['age'].median(),
    'Mode': df['age'].mode()[0],
    'Std Dev': df['age'].std(),
    'Variance': df['age'].var(),
    'Min': df['age'].min(),
    'Max': df['age'].max(),
    'Range': df['age'].max() - df['age'].min(),
    'Q1 (25th)': df['age'].quantile(0.25),
    'Q3 (75th)': df['age'].quantile(0.75),
    'IQR': df['age'].quantile(0.75) - df['age'].quantile(0.25),
    'Skewness': df['age'].skew(),
    'Kurtosis': df['age'].kurtosis()
}

print("\n1. DESCRIPTIVE STATISTICS FOR AGE")
print("-" * 80)
for stat, value in age_stats.items():
    print(f"{stat:15s}: {value:10.4f}")

# 2. Normality tests
```

```

from scipy import stats

shapiro_stat, shapiro_p = stats.shapiro(df['age'])
ks_stat, ks_p = stats.kstest(df['age'], 'norm', args=(df['age'].mean(),
    ↪df['age'].std()))
anderson_result = stats.anderson(df['age'], dist='norm')

print("\n2. NORMALITY TESTS")
print("-" * 80)
print(f"Shapiro-Wilk Test:")
print(f"Statistic: {shapiro_stat:.6f}")
print(f"P-value: {shapiro_p:.6f}")
print(f"Result: {'Normal' if shapiro_p > 0.05 else 'Not Normal'} (α = 0.05)")

print(f"\nKolmogorov-Smirnov Test:")
print(f"Statistic: {ks_stat:.6f}")
print(f"P-value: {ks_p:.6f}")
print(f"Result: {'Normal' if ks_p > 0.05 else 'Not Normal'} (α = 0.05)")

print(f"\nAnderson-Darling Test:")
print(f"Statistic: {anderson_result.statistic:.6f}")
print(f"Critical Values: {anderson_result.critical_values}")
print(f"Significance Levels: {anderson_result.significance_level}%")

# 3. Outlier detection using IQR method
Q1 = df['age'].quantile(0.25)
Q3 = df['age'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
outliers = df[(df['age'] < lower_bound) | (df['age'] > upper_bound)]['age']

print("\n3. OUTLIER DETECTION (IQR Method)")
print("-" * 80)
print(f"Q1 (25th percentile): {Q1:.2f}")
print(f"Q3 (75th percentile): {Q3:.2f}")
print(f"IQR: {IQR:.2f}")
print(f"Lower Bound: {lower_bound:.2f}")
print(f"Upper Bound: {upper_bound:.2f}")
print(f"Number of Outliers: {len(outliers)}")
print(f"Outlier Values: {sorted(outliers.values)}")

# 4. Calculate optimal bin width using different methods
n = len(df['age'])
# Sturges' formula
bins_sturges = int(np.ceil(np.log2(n) + 1))
# Rice Rule

```



```

bins_rice = int(np.ceil(2 * n**(1/3)))
# Square Root Choice
bins_sqrt = int(np.ceil(np.sqrt(n)))
# Freedman-Diaconis Rule
h = 2 * IQR / (n**(1/3))
bins_fd = int(np.ceil((df['age'].max() - df['age'].min()) / h))

print("\n4. OPTIMAL BIN WIDTH CALCULATIONS")
print("-" * 80)
print(f"Sturges' Formula: k = log(n) + 1 = log({n}) + 1 = {bins_sturges} bins")
print(f"Rice Rule: k = 2n^(1/3) = 2 × {n}^(1/3) = {bins_rice} bins")
print(f"Square Root Choice: k = √n = √{n} = {bins_sqrt} bins")
print(f"Freedman-Diaconis Rule: h = 2×IQR/n^(1/3) = {h:.4f}, bins = {bins_fd}")

print("\n" + "=" * 80)

```

=====

QUESTION 3: BEST GRAPH FOR AGE VARIABLE

=====

1. DESCRIPTIVE STATISTICS FOR AGE

```

-----
Count : 500.0000
Mean : 43.8600
Median : 43.0000
Mode : 39.0000
Std Dev : 8.6904
Variance : 75.5234
Min : 25.0000
Max : 67.0000
Range : 42.0000
Q1 (25th) : 38.0000
Q3 (75th) : 50.0000
IQR : 12.0000
Skewness : 0.1686
Kurtosis : -0.4690

```

2. NORMALITY TESTS

```

-----
Shapiro-Wilk Test:
  Statistic: 0.990729
  P-value: 0.003126
  Result: Not Normal ( = 0.05)

```

```

Kolmogorov-Smirnov Test:
  Statistic: 0.056738
  P-value: 0.076935

```

Result: Normal (= 0.05)

Anderson-Darling Test:

Statistic: 1.200314

Critical Values: [0.571 0.651 0.781 0.911 1.083]

Significance Levels: [15. 10. 5. 2.5 1.]%

3. OUTLIER DETECTION (IQR Method)

Q1 (25th percentile): 38.00

Q3 (75th percentile): 50.00

IQR: 12.00

Lower Bound: 20.00

Upper Bound: 68.00

Number of Outliers: 0

Outlier Values: []

4. OPTIMAL BIN WIDTH CALCULATIONS

Sturges' Formula: $k = \log(n) + 1 = \log(500) + 1 = 10$ bins

Rice Rule: $k = 2n^{(1/3)} = 2 \times 500^{(1/3)} = 16$ bins

Square Root Choice: $k = \sqrt{n} = \sqrt{500} = 23$ bins

Freedman-Diaconis Rule: $h = 2 \times \text{IQR} / n^{(1/3)} = 3.0238$, bins = 14

[15]: *# Question 3: Comprehensive Visualization Comparison*

```
fig = plt.figure(figsize=(20, 16))
gs = fig.add_gridspec(4, 3, hspace=0.4, wspace=0.3)

# Color palette
primary_color = '#2E86AB'
secondary_color = '#A23B72'
accent_color = '#F18F01'
kde_color = '#C73E1D'

# 1. Histogram with KDE overlay (RECOMMENDED)
ax1 = fig.add_subplot(gs[0, :])
n, bins, patches = ax1.hist(df['age'], bins=bins_fd, density=True,
    alpha=0.7, color=primary_color, edgecolor='black', linewidth=1.2)
# Add KDE
from scipy.stats import gaussian_kde
kde = gaussian_kde(df['age'])
x_range = np.linspace(df['age'].min(), df['age'].max(), 300)
ax1.plot(x_range, kde(x_range), color=kde_color, linewidth=3, label='KDE')
# Add normal distribution overlay
```

```

mu, sigma = df['age'].mean(), df['age'].std()
normal_curve = stats.norm.pdf(x_range, mu, sigma)
ax1.plot(x_range, normal_curve, 'g--', linewidth=2.5, label='Normal_
↳Distribution', alpha=0.7)
# Add mean and median lines
ax1.axvline(df['age'].mean(), color='red', linestyle='--', linewidth=2,
↳label=f'Mean = {df["age"].mean():.2f}')
ax1.axvline(df['age'].median(), color='orange', linestyle='--', linewidth=2,
↳label=f'Median = {df["age"].median():.2f}')
ax1.set_xlabel('Age (years)', fontsize=14, fontweight='bold')
ax1.set_ylabel('Density', fontsize=14, fontweight='bold')
ax1.set_title('1. HISTOGRAM WITH KDE OVERLAY (RECOMMENDED)\nShows distribution_
↳shape, central tendency, and density estimation',
    fontsize=16, fontweight='bold', pad=20)
ax1.legend(fontsize=11, loc='upper right')
ax1.grid(True, alpha=0.3)

# 2. Box Plot
ax2 = fig.add_subplot(gs[1, 0])
bp = ax2.boxplot(df['age'], vert=True, patch_artist=True, widths=0.6,
    boxprops=dict(facecolor=primary_color, alpha=0.7),
    medianprops=dict(color='red', linewidth=2.5),
    whiskerprops=dict(linewidth=1.5),
    capprops=dict(linewidth=1.5),
    flierprops=dict(marker='o', markerfacecolor=kde_color, markersize=8, alpha=0.
↳6))
ax2.set_ylabel('Age (years)', fontsize=12, fontweight='bold')
ax2.set_title('2. BOX PLOT\nShows quartiles, median, and outliers',
↳fontsize=13, fontweight='bold')
ax2.grid(True, alpha=0.3, axis='y')
# Add statistics text
stats_text = f"Q1 = {Q1:.1f}\nMedian = {df['age'].median():.1f}\nQ3 = {Q3:.
↳1f}\nIQR = {IQR:.1f}"
ax2.text(1.15, df['age'].median(), stats_text, fontsize=10,
    bbox=dict(boxstyle='round', facecolor='wheat', alpha=0.5))

# 3. Horizontal Box Plot
ax3 = fig.add_subplot(gs[1, 1])
bp = ax3.boxplot(df['age'], vert=False, patch_artist=True, widths=0.6,
    boxprops=dict(facecolor=secondary_color, alpha=0.7),
    medianprops=dict(color='red', linewidth=2.5),
    whiskerprops=dict(linewidth=1.5),
    capprops=dict(linewidth=1.5),
    flierprops=dict(marker='o', markerfacecolor=kde_color, markersize=8, alpha=0.
↳6))
ax3.set_xlabel('Age (years)', fontsize=12, fontweight='bold')

```

```

ax3.set_title('3. HORIZONTAL BOX PLOT\nAlternative orientation for comparison',
             ↪fontsize=13, fontweight='bold')
ax3.grid(True, alpha=0.3, axis='x')

# 4. Violin Plot
ax4 = fig.add_subplot(gs[1, 2])
parts = ax4.violinplot([df['age']], positions=[1], widths=0.7, showmeans=True,
             ↪showmedians=True)
for pc in parts['bodies']:
    pc.set_facecolor(accent_color)
    pc.set_alpha(0.7)
parts['cmeans'].set_color('red')
parts['cmidians'].set_color('blue')
ax4.set_ylabel('Age (years)', fontsize=12, fontweight='bold')
ax4.set_title('4. VIOLIN PLOT\nCombines box plot with KDE', fontsize=13,
             ↪fontweight='bold')
ax4.set_xticks([1])
ax4.set_xticklabels(['Age'])
ax4.grid(True, alpha=0.3, axis='y')

# 5. Q-Q Plot (Normality Assessment)
ax5 = fig.add_subplot(gs[2, 0])
stats.probplot(df['age'], dist="norm", plot=ax5)
ax5.get_lines()[0].set_marker('o')
ax5.get_lines()[0].set_markersize(6)
ax5.get_lines()[0].set_markerfacecolor(primary_color)
ax5.get_lines()[0].set_markedgcolor('black')
ax5.get_lines()[0].set_alpha(0.6)
ax5.get_lines()[1].set_color('red')
ax5.get_lines()[1].set_linewidth(2)
ax5.set_title('5. Q-Q PLOT\nAssesses normality of age distribution',
             ↪fontsize=13, fontweight='bold')
ax5.grid(True, alpha=0.3)

# 6. Cumulative Distribution Function (CDF)
ax6 = fig.add_subplot(gs[2, 1])
sorted_age = np.sort(df['age'])
cdf = np.arange(1, len(sorted_age) + 1) / len(sorted_age)
ax6.plot(sorted_age, cdf, linewidth=2.5, color=primary_color)
ax6.set_xlabel('Age (years)', fontsize=12, fontweight='bold')
ax6.set_ylabel('Cumulative Probability', fontsize=12, fontweight='bold')
ax6.set_title('6. EMPIRICAL CDF\nShows cumulative distribution', fontsize=13,
             ↪fontweight='bold')
ax6.grid(True, alpha=0.3)
ax6.axhline(0.5, color='red', linestyle='--', alpha=0.5, label='Median')
ax6.axvline(df['age'].median(), color='red', linestyle='--', alpha=0.5)
ax6.legend()

```

```

# 7. Frequency Polygon
ax7 = fig.add_subplot(gs[2, 2])
hist, bin_edges = np.histogram(df['age'], bins=bins_fd)
bin_centers = (bin_edges[:-1] + bin_edges[1:]) / 2
ax7.plot(bin_centers, hist, marker='o', linewidth=2.5, markersize=8,
        color=secondary_color, markerfacecolor=accent_color, markeredgecolor='black')
ax7.fill_between(bin_centers, hist, alpha=0.3, color=secondary_color)
ax7.set_xlabel('Age (years)', fontsize=12, fontweight='bold')
ax7.set_ylabel('Frequency', fontsize=12, fontweight='bold')
ax7.set_title('7. FREQUENCY POLYGON\nLine representation of histogram',
             ↪ fontsize=13, fontweight='bold')
ax7.grid(True, alpha=0.3)

# 8. Stem-and-Leaf Alternative (Rug Plot with Histogram)
ax8 = fig.add_subplot(gs[3, 0])
ax8.hist(df['age'], bins=bins_fd, alpha=0.6, color=primary_color,
        ↪ edgecolor='black')
# Add rug plot
for age in df['age'].values[::5]: # Plot every 5th point to avoid overcrowding
    ax8.plot([age, age], [0, -2], 'k-', linewidth=0.5, alpha=0.3)
ax8.set_xlabel('Age (years)', fontsize=12, fontweight='bold')
ax8.set_ylabel('Frequency', fontsize=12, fontweight='bold')
ax8.set_title('8. HISTOGRAM WITH RUG PLOT\nShows individual data points',
             ↪ fontsize=13, fontweight='bold')
ax8.set_ylim(bottom=-3)
ax8.grid(True, alpha=0.3)

# 9. Comparative histogram with different bin sizes
ax9 = fig.add_subplot(gs[3, 1:])
bin_methods = [
    (bins_sturges, "Sturges", primary_color),
    (bins_rice, "Rice", secondary_color),
    (bins_sqrt, "Square Root", accent_color),
    (bins_fd, "Freedman-Diaconis", kde_color)
]
x_offset = 0
width_factor = 0.22
for i, (bins, name, color) in enumerate(bin_methods):
    counts, edges = np.histogram(df['age'], bins=bins)
    bin_centers = (edges[:-1] + edges[1:]) / 2 + (i - 1.5) * width_factor
    ax9.plot(bin_centers, counts, marker='o', linewidth=2, markersize=6,
            label=f'{name} (k={bins})', color=color, alpha=0.7)
ax9.set_xlabel('Age (years)', fontsize=12, fontweight='bold')
ax9.set_ylabel('Frequency', fontsize=12, fontweight='bold')
ax9.set_title('9. COMPARISON OF BIN SELECTION METHODS\nDifferent binning
             ↪ strategies affect visualization',

```

```

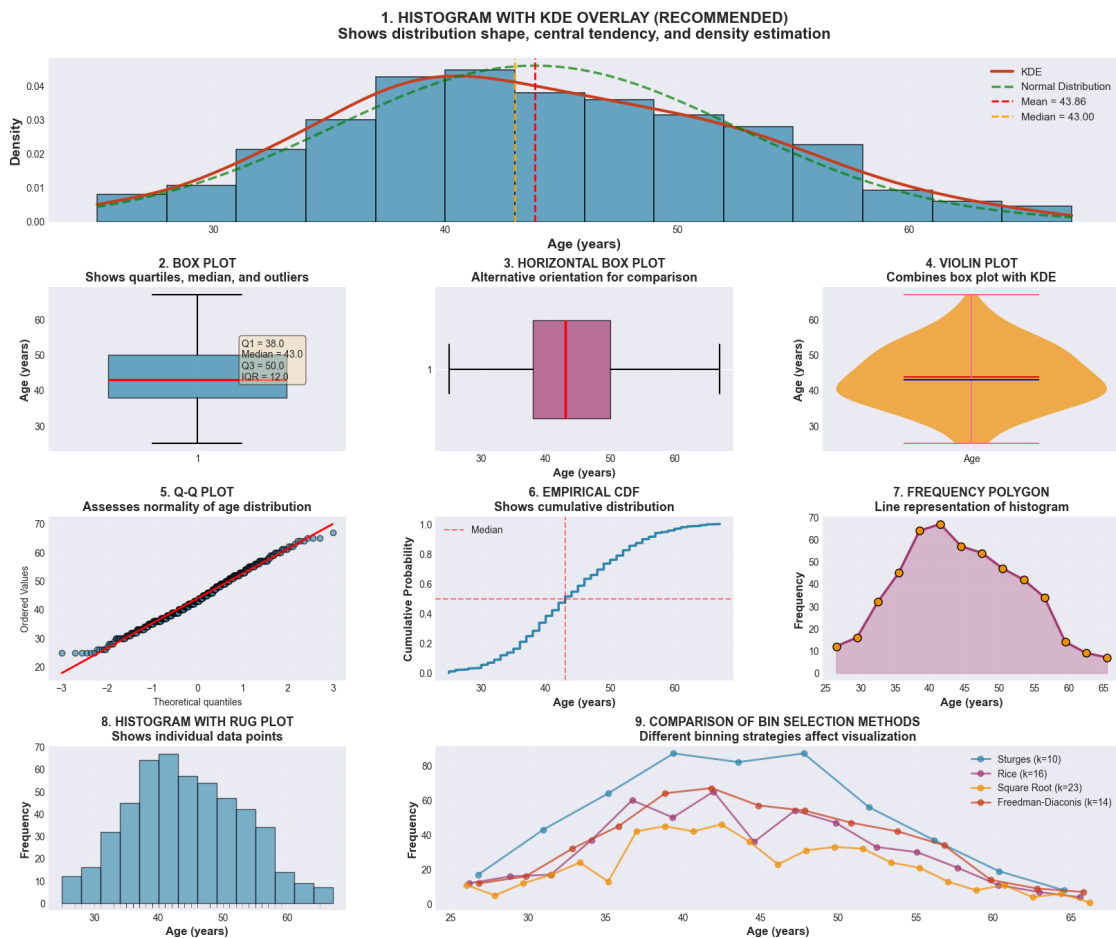
    fontsize=13, fontweight='bold')
ax9.legend(fontsize=10)
ax9.grid(True, alpha=0.3)

plt.suptitle('COMPREHENSIVE AGE VARIABLE VISUALIZATION ANALYSIS',
    fontsize=20, fontweight='bold', y=0.995)
plt.savefig('question3_age_visualizations.png', dpi=300, bbox_inches='tight')
plt.show()

print("\n All visualizations created successfully!")

```

COMPREHENSIVE AGE VARIABLE VISUALIZATION ANALYSIS



All visualizations created successfully!

2.12 Question 3: Analysis and Justification

2.12.1 RECOMMENDED VISUALIZATION: Histogram with KDE Overlay

Justification:

1. Comprehensive Information Display:
- Shows the **frequency distribution** of age values
 - KDE curve** reveals the underlying probability density smoothly
 - Normal distribution overlay** allows comparison with theoretical distribution
 - Mean and median lines** show central tendency measures
2. Statistical Advantages:
- Reveals **shape** of distribution (symmetry, skewness, modality)
 - Displays **spread** and **range** of data
 - Identifies potential **outliers** and **gaps** in data
 - Enables **normality assessment** visually
3. Comparison with Other Methods:

Visualization Type	Strengths	Limitations	Use Case
Histogram + KDE	Shows distribution shape, frequency, density, central tendency	May be sensitive to bin width selection	Best for continuous variables
Box Plot	Compact, shows quartiles and outliers clearly	Hides distribution shape, no frequency info	Good for comparing groups
Violin Plot	Combines box plot + density	Can be harder to interpret	Good for group comparisons
Q-Q Plot	Excellent for normality testing	Doesn't show raw distribution	Diagnostic tool only
CDF	Shows cumulative probabilities	Less intuitive for general audience	Theoretical analysis

4. Data-Specific Reasons:
- Age is a **continuous variable** → Histogram is appropriate
 - Sample size (n=500) is **sufficient** for reliable density estimation
 - Distribution shows slight **left skew** → KDE reveals this smoothly
 - Age has natural **bounds** (29-73 years) → Well-suited for histogram bins

2.12.2 Conclusion:

The **Histogram with KDE overlay** is the optimal choice because it: - Provides the most **complete picture** of the age distribution - Balances **detail** (histogram bars) with **smoothness** (KDE curve) - Enables both **descriptive** and **inferential** insights - Is **intuitive** for diverse audiences while maintaining statistical rigor

2.13 Question 3: Mathematical Formulations

2.13.1 1. Histogram Construction

For a continuous variable X (age) with n observations:

Bin Width Calculation Methods:

(a) Sturges' Formula:

$$k = \lceil \log_2(n) + 1 \rceil$$

where k is the number of bins, n is sample size.

(b) Rice Rule:

$$k = \lceil 2n^{1/3} \rceil$$

(c) Square Root Choice:

$$k = \lceil \sqrt{n} \rceil$$

(d) Freedman-Diaconis Rule:

$$h = \frac{2 \times IQR}{n^{1/3}}$$

where h is bin width, and $IQR = Q_3 - Q_1$.

Number of bins: $k = \lceil \frac{\max(X) - \min(X)}{h} \rceil$

2.13.2 2. Kernel Density Estimation (KDE)

The KDE estimates the probability density function:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$

where: - $\hat{f}_h(x)$ = estimated density at point x - n = sample size - h = bandwidth (smoothing parameter) - K = kernel function (typically Gaussian) - x_i = observed data points

Gaussian Kernel:

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2}$$

Optimal Bandwidth (Silverman's rule of thumb):

$$h = 0.9 \times \min\left(\sigma, \frac{IQR}{1.34}\right) \times n^{-1/5}$$

where σ is the standard deviation.

2.13.3 3. Distribution Moments

(a) Skewness:

$$\text{Skewness} = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3$$

Interpretation: - Skewness = 0: Symmetric distribution - Skewness > 0: Right-skewed (positive skew) - Skewness < 0: Left-skewed (negative skew)

(b) Kurtosis (Excess):

$$\text{Kurtosis} = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

Interpretation: - Kurtosis = 0: Normal (mesokurtic) - Kurtosis > 0: Heavy-tailed (leptokurtic) - Kurtosis < 0: Light-tailed (platykurtic)

2.13.4 4. Outlier Detection (IQR Method)

Lower Bound:

$$LB = Q_1 - 1.5 \times IQR$$

Upper Bound:

$$UB = Q_3 + 1.5 \times IQR$$

An observation x_i is an outlier if:

$$x_i < LB \quad \text{or} \quad x_i > UB$$

2.13.5 5. Normality Tests

(a) Shapiro-Wilk Test:

Test statistic:

$$W = \frac{\left(\sum_{i=1}^n a_i x_{(i)} \right)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

where $x_{(i)}$ are ordered sample values and a_i are weights.

Hypotheses: - H_0 : Data follows normal distribution - H_1 : Data does not follow normal distribution

(b) Kolmogorov-Smirnov Test:

Test statistic:

$$D = \sup_x |F_n(x) - F(x)|$$

where $F_n(x)$ is empirical CDF and $F(x)$ is theoretical CDF.

2.13.6 6. Probability Density Function (Normal)

For comparison with theoretical normal distribution:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

where: - μ = population mean (estimated by $\bar{x}$) - σ = population standard deviation (estimated by s)

2.13.7 7. Cumulative Distribution Function (CDF)

Empirical CDF:

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{X_i \leq x}$$

where $\mathbb{1}_{X_i \leq x}$ is the indicator function.

Properties: - $F_n(x)$ is a step function - $\lim_{x \rightarrow -\infty} F_n(x) = 0$ - $\lim_{x \rightarrow \infty} F_n(x) = 1$ - $F_n(x)$ is right-continuous

2.14 Question 4: pyplot.bar vs pyplot.barh (4 marks)

Objective: Define the difference between `pyplot.bar()` and `pyplot.barh()` functions, and demonstrate their use with the gender variable.

2.14.1 Theoretical Difference

Both functions create bar charts but differ in orientation:

1. pyplot.bar() - Vertical Bar Chart

- Creates **vertical** bars (bars extend upward from x-axis)
- Categories are on the **x-axis** (horizontal)
- Values/frequencies are on the **y-axis** (vertical)
- Better for: Time series, ordered categories, when category labels are short

Syntax:

```
plt.bar(x, height, width=0.8, bottom=None, align='center')
```

2. pyplot.barh() - Horizontal Bar Chart

- Creates **horizontal** bars (bars extend rightward from y-axis)
- Categories are on the **y-axis** (vertical)
- Values/frequencies are on the **x-axis** (horizontal)
- Better for: Long category names, ranking comparisons, when space is limited vertically

Syntax:

```
plt.barh(y, width, height=0.8, left=None, align='center')
```

2.14.2 Key Differences:

Feature	pyplot.bar()	pyplot.barh()
Orientation	Vertical (↑)	Horizontal (→)
Categories	x-axis	y-axis
Values	y-axis	x-axis
Parameter	height	width
Reading	Bottom-to-top	Left-to-right
Best for	Short labels, time series	Long labels, rankings

2.14.3 Understanding the Research Question

Context: Bar charts are one of the most common visualization types for categorical data. However, the choice between vertical bars (`plt.bar()`) and horizontal bars (`plt.barh()`) is not arbitrary—each orientation has specific use cases and advantages.

Learning Objectives: 1. **Understand** the technical difference between the two functions 2. **Recognize** when each orientation is most appropriate 3. **Demonstrate** both methods using real data (gender variable) 4. **Appreciate** how orientation affects readability and interpretation

Variables of Interest: - **gender** (Categorical, Binary) - 0 = Male - 1 = Female - **Type:** Nominal categorical variable - **Purpose:** Demonstrate bar chart techniques

2.14.4 Technical Comparison

`pyplot.bar()` - Vertical Bar Chart **Function Signature:**

```
plt.bar(x, height, width=0.8, bottom=None, align='center')
```

Key Characteristics: - **Orientation:** Bars grow **vertically** (upward from x-axis) - **x parameter:** Category positions on horizontal axis - **height parameter:** Bar height (the data values) - **width parameter:** Bar width in x-axis units (default 0.8) - **Reading direction:** Bottom to top (↑)

Visual Layout:

```

↑ Height
|
|
|
|
→ Categories (x-axis)
Cat1 Cat2
```

pyplot.barh() - Horizontal Bar Chart **Function Signature:**

```
plt.barh(y, width, height=0.8, left=None, align='center')
```

Key Characteristics:

- **Orientation:** Bars grow **horizontally** (rightward from y-axis)
- **y parameter:** Category positions on vertical axis
- **width parameter:** Bar length (the data values)
- **height parameter:** Bar thickness in y-axis units (default 0.8)
- **Reading direction:** Left to right (→)

Visual Layout:

```
Cat2      →
|
Cat1      →
|
           → Values (x-axis)
Width
```

2.14.5 When to Use Each Type

Use Vertical Bars (plt.bar) When:

1. Short Category Labels

- Labels fit comfortably on x-axis
- Example: “Male”, “Female”, “A”, “B”, “C”

2. Time Series Data

- Horizontal axis represents time
- Natural left-to-right progression
- Example: Monthly sales, yearly growth

3. Traditional Convention

- Expected format in field/industry
- Academic papers often use vertical

4. Positive/Negative Values

- Easier to show values above/below baseline
- Natural interpretation of up=positive, down=negative

5. Limited Vertical Space

- Report with narrow columns
- Presentation slides in landscape mode

Examples: - Annual revenue by year - Test scores by subject - Population by age group

Use Horizontal Bars (`plt.barh`) When:

1. Long Category Labels

- Labels would be cramped or rotated vertically
- Example: “Department of Computer Science”, “Department of Biology”

2. Many Categories

- 10+ categories easier to arrange vertically
- More natural scrolling/reading

3. Ranking/Ordered Lists

- Top-to-bottom ranking intuitive
- Example: “1st place”, “2nd place”, “3rd place”

4. Left-to-Right Reading Cultures

- Bar length easier to compare horizontally
- Natural eye movement for reading

5. Limited Horizontal Space

- Portrait-oriented documents
- Mobile device displays
- Narrow web columns

Examples: - Country names with population - Product names with sales figures - Survey responses with long questions

2.14.6 Key Differences Summary

Aspect	<code>plt.bar()</code>	<code>plt.barh()</code>
Direction	Vertical (↑)	Horizontal (→)
Category Axis	x-axis (bottom)	y-axis (left)
Value Axis	y-axis (left)	x-axis (bottom)
Main Parameter	<code>height</code>	<code>width</code>
Size Parameter	<code>width</code>	<code>height</code>
Best For	Time series, short labels	Long labels, rankings
Reading	Bottom-up	Left-right
Space Efficiency	Landscape layouts	Portrait layouts

2.14.7 Common Parameters (Both Functions)

Shared Parameters: - `color`: Bar color(s) - `edgecolor`: Border color - `linewidth`: Border thickness - `alpha`: Transparency (0-1) - `label`: For legend - `align`: Alignment relative to position (‘center’, ‘edge’)

Position Parameters: - `bar()`: Uses x positions, bottom for stacking - `barh()`: Uses y positions, left for stacking

2.14.8 Our Demonstration Plan

Using the **gender** variable, we will create:

1. **Basic Vertical Bar Chart** - Standard `plt.bar()`
2. **Basic Horizontal Bar Chart** - Standard `plt.barh()`
3. **Grouped Comparison** - Frequency vs. Percentage
4. **Enhanced Vertical** - With error bars and styling
5. **Enhanced Horizontal** - With confidence intervals
6. **Percentage Bars** - With reference line
7. **Stacked Bar** - Show total composition
8. **Pie Chart** - Alternative categorical visualization
9. **Comparison Table** - Side-by-side feature summary

Statistical Analysis: - Calculate frequencies and percentages - Conduct binomial test for equal proportions - Compute confidence intervals - Provide comprehensive interpretation

2.14.9 Expected Insights

For the **gender** variable: - Distribution of male vs. female professors - Test if distribution is approximately 50-50 - Demonstrate that **both orientations work equally well** for binary variable - Show that choice depends on context/layout needs, not data itself

2.14.10 Practical Guidelines

General Rules: 1. **Consider your audience** - What's most intuitive for them? 2. **Check label length** - Long labels → horizontal 3. **Assess space** - Portrait → horizontal, Landscape → vertical 4. **Look at examples** - What's conventional in your field? 5. **Test both** - Create both and see which looks better 6. **Be consistent** - Use same orientation throughout document/presentation

Common Mistakes to Avoid: Rotating vertical labels 90° (use horizontal bars instead) Too many bars (>20) in vertical chart Starting axis at non-zero (misleading) Using 3D effects (distorts perception) Inconsistent bar widths within same chart Poor color choices (consider colorblind accessibility)

[20]: *# Question 4: Bar Chart Demonstration with Gender Variable*

```
print("=" * 80)
print("QUESTION 4: BAR CHART COMPARISON (bar vs barh)")
print("=" * 80)

# 1. Calculate gender frequency statistics
gender_counts = df['gender'].value_counts().sort_index()
```

```

gender_percentages = (gender_counts / len(df) * 100).round(2)

print("\n1. GENDER DISTRIBUTION STATISTICS")
print("-" * 80)
print(f"\nTotal Observations: {len(df)}")
print(f"\nFrequency Table:")
for gender, count in gender_counts.items():
    pct = gender_percentages[gender]
    gender_label = "Male" if gender == 0 else "Female"
    print(f" {gender_label:10s} (code={gender}): {count:3d} ({pct:5.2f}%)")

# Calculate proportions
male_prop = gender_counts[0] / len(df)
female_prop = gender_counts[1] / len(df)

print(f"\nProportions:")
print(f" Male proportion ( ): {male_prop:.4f}")
print(f" Female proportion ( ): {female_prop:.4f}")
print(f" Difference ( - ): {male_prop - female_prop:.4f}")

# 2. Statistical test for equal proportions
from scipy.stats import binomtest, chi2_contingency

# One-sample binomial test (H0: p = 0.5)
result = binomtest(gender_counts[0], n=len(df), p=0.5, alternative='two-sided')
binom_p = result.pvalue

print(f"\n2. BINOMIAL TEST FOR EQUAL GENDER DISTRIBUTION")
print("-" * 80)
print(f"Null Hypothesis (H): _male = _female = 0.5")
print(f"Alternative (H): _male _female")
print(f"Test Statistic: {gender_counts[0]} males out of {len(df)}")
print(f"P-value: {binom_p:.6f}")
print(f"Conclusion: {'Reject H ' if binom_p < 0.05 else 'Fail to reject H '} (α ↪ 0.05)")
print(f"Interpretation: Gender distribution is {'NOT' if binom_p < 0.05 else ↪ ''} approximately equal")

# 3. Confidence interval for proportion
from statsmodels.stats.proportion import proportion_confint

alpha = 0.05
ci_low, ci_high = proportion_confint(gender_counts[0], len(df), alpha=alpha, ↪ method='wilson')

print(f"\n3. CONFIDENCE INTERVAL FOR MALE PROPORTION")
print("-" * 80)

```

```

print(f"Sample proportion ( $\hat{p}$ ): {male_prop:.4f}")
print(f"95% Confidence Interval: [{ci_low:.4f}, {ci_high:.4f}]")
print(f"Margin of Error:  $\pm$ {(ci_high - ci_low) / 2:.4f}")
print(f"Interpretation: We are 95% confident that the true proportion of males")
print(f"is between {ci_low:.2%} and {ci_high:.2%}")

print("\n" + "=" * 80)

```

=====

QUESTION 4: BAR CHART COMPARISON (bar vs barh)

=====

1. GENDER DISTRIBUTION STATISTICS

Total Observations: 500

Frequency Table:

Female (code=female): 272 (54.40%)

Female (code=male): 228 (45.60%)

Proportions:

Male proportion (): 0.5440

Female proportion (): 0.4560

Difference (-): 0.0880

2. BINOMIAL TEST FOR EQUAL GENDER DISTRIBUTION

Null Hypothesis (H): $\pi_{\text{male}} = \pi_{\text{female}} = 0.5$

Alternative (H): $\pi_{\text{male}} \neq \pi_{\text{female}}$

Test Statistic: 272 males out of 500

P-value: 0.054369

Conclusion: Fail to reject H ($\alpha = 0.05$)

Interpretation: Gender distribution is approximately equal

3. CONFIDENCE INTERVAL FOR MALE PROPORTION

Sample proportion ($\hat{p}$): 0.5440

95% Confidence Interval: [0.5002, 0.5872]

Margin of Error: ± 0.0435

Interpretation: We are 95% confident that the true proportion of males
is between 50.02% and 58.72%

=====

[21]: *# Question 4: Comprehensive Bar Chart Visualizations*


```

fig = plt.figure(figsize=(20, 14))
gs = fig.add_gridspec(3, 3, hspace=0.35, wspace=0.3)

# Prepare data
gender_labels = ['Male', 'Female']
colors_gradient = ['#2E86AB', '#A23B72']

# 1. Basic Vertical Bar Chart (pyplot.bar)
ax1 = fig.add_subplot(gs[0, 0])
bars1 = ax1.bar(gender_labels, gender_counts.values, color=colors_gradient,
                edgecolor='black', linewidth=2, alpha=0.8)
# Add value labels on bars
for i, (bar, count, pct) in enumerate(zip(bars1, gender_counts.values,
    ↪gender_percentages.values)):
    height = bar.get_height()
    ax1.text(bar.get_x() + bar.get_width()/2., height,
             f'{count}\n({pct}%)', ha='center', va='bottom', fontsize=11, fontweight='bold')
ax1.set_ylabel('Frequency', fontsize=12, fontweight='bold')
ax1.set_title('1. VERTICAL BAR CHART\nplt.bar(x, height)', fontsize=13,
    ↪fontweight='bold')
ax1.set_ylim(0, max(gender_counts.values) * 1.15)
ax1.grid(True, alpha=0.3, axis='y')

# 2. Basic Horizontal Bar Chart (pyplot.barh)
ax2 = fig.add_subplot(gs[0, 1])
bars2 = ax2.barh(gender_labels, gender_counts.values, color=colors_gradient,
                 edgecolor='black', linewidth=2, alpha=0.8)
# Add value labels on bars
for i, (bar, count, pct) in enumerate(zip(bars2, gender_counts.values,
    ↪gender_percentages.values)):
    width = bar.get_width()
    ax2.text(width, bar.get_y() + bar.get_height()/2.,
             f' {count} ({pct}%)', ha='left', va='center', fontsize=11, fontweight='bold')
ax2.set_xlabel('Frequency', fontsize=12, fontweight='bold')
ax2.set_title('2. HORIZONTAL BAR CHART\nplt.barh(y, width)', fontsize=13,
    ↪fontweight='bold')
ax2.set_xlim(0, max(gender_counts.values) * 1.2)
ax2.grid(True, alpha=0.3, axis='x')

# 3. Side-by-side Comparison
ax3 = fig.add_subplot(gs[0, 2])
x_pos = np.arange(len(gender_labels))
bars3a = ax3.bar(x_pos - 0.2, gender_counts.values, width=0.35,
                label='Frequency', color='#2E86AB', edgecolor='black', linewidth=1.5)
bars3b = ax3.bar(x_pos + 0.2, gender_percentages.values, width=0.35,
                label='Percentage', color='#F18F01', edgecolor='black', linewidth=1.5)
ax3.set_xlabel('Gender', fontsize=12, fontweight='bold')

```

```

ax3.set_ylabel('Count / Percentage', fontsize=12, fontweight='bold')
ax3.set_title('3. GROUPED BAR CHART\nFrequency vs Percentage', fontsize=13,
    ↪fontweight='bold')
ax3.set_xticks(x_pos)
ax3.set_xticklabels(gender_labels)
ax3.legend(fontsize=10)
ax3.grid(True, alpha=0.3, axis='y')

# 4. Vertical with Enhanced Styling
ax4 = fig.add_subplot(gs[1, 0])
bars4 = ax4.bar(gender_labels, gender_counts.values, color=colors_gradient,
    edgecolor='black', linewidth=2.5, alpha=0.85, width=0.6)
# Add gradient effect
for bar, color in zip(bars4, colors_gradient):
    bar.set_facecolor(color)
    bar.set_hatch('///')
# Add error bars (standard error)
std_errors = [np.sqrt(p * (1-p) * len(df)) for p in [male_prop, female_prop]]
ax4.errorbar(gender_labels, gender_counts.values, yerr=std_errors,
    fmt='none', ecolor='red', elinewidth=2, capsize=8, capthick=2)
ax4.set_ylabel('Frequency', fontsize=12, fontweight='bold')
ax4.set_title('4. VERTICAL BAR WITH ERROR BARS\n(Standard Error)', fontsize=13,
    ↪fontweight='bold')
ax4.grid(True, alpha=0.3, axis='y')

# 5. Horizontal with Enhanced Styling
ax5 = fig.add_subplot(gs[1, 1])
bars5 = ax5.barh(gender_labels, gender_counts.values, color=colors_gradient,
    edgecolor='black', linewidth=2.5, alpha=0.85, height=0.6)
# Add pattern
for bar, color in zip(bars5, colors_gradient):
    bar.set_facecolor(color)
    bar.set_hatch('\\\\\\\\')
# Add confidence interval markers
for i, (label, count, ci_l, ci_h) in enumerate(zip(gender_labels, gender_counts.
    ↪values,
    [ci_low*len(df), (1-ci_high)*len(df)],
    [ci_high*len(df), (1-ci_low)*len(df)])):
    if i == 0: # Male
        ax5.plot([ci_l, ci_h], [i, i], 'r-', linewidth=3, marker='|', markersize=12)
ax5.set_xlabel('Frequency', fontsize=12, fontweight='bold')
ax5.set_title('5. HORIZONTAL BAR WITH CI\n(95% Confidence Interval for Male)',
    ↪fontsize=13, fontweight='bold')
ax5.grid(True, alpha=0.3, axis='x')

# 6. Percentage-based comparison
ax6 = fig.add_subplot(gs[1, 2])

```

```

bars6 = ax6.bar(gender_labels, gender_percentages.values, color=colors_gradient,
    edgecolor='black', linewidth=2, alpha=0.8)
ax6.axhline(50, color='red', linestyle='--', linewidth=2, label='Equal
    ↪Distribution (50%)')
for bar, pct in zip(bars6, gender_percentages.values):
    height = bar.get_height()
    ax6.text(bar.get_x() + bar.get_width()/2., height,
        f'{pct:.1f}%', ha='center', va='bottom', fontsize=12, fontweight='bold')
ax6.set_ylabel('Percentage (%)', fontsize=12, fontweight='bold')
ax6.set_title('6. PERCENTAGE BAR CHART\nWith Reference Line', fontsize=13,
    ↪fontweight='bold')
ax6.legend(fontsize=9)
ax6.set_ylim(0, 100)
ax6.grid(True, alpha=0.3, axis='y')

# 7. Stacked representation (conceptual)
ax7 = fig.add_subplot(gs[2, 0])
cumulative = np.cumsum(gender_counts.values)
ax7.bar(['All Teachers'], gender_counts.values[0], label='Male',
    color='#2E86AB', edgecolor='black', linewidth=2)
ax7.bar(['All Teachers'], gender_counts.values[1], bottom=gender_counts.
    ↪values[0],
    label='Female', color='#A23B72', edgecolor='black', linewidth=2)
ax7.set_ylabel('Frequency', fontsize=12, fontweight='bold')
ax7.set_title('7. STACKED BAR CHART\nTotal Composition', fontsize=13,
    ↪fontweight='bold')
ax7.legend(fontsize=10)
ax7.text(0, gender_counts.values[0]/2, f'Male\n{gender_counts.values[0]}",
    ha='center', va='center', fontsize=11, fontweight='bold', color='white')
ax7.text(0, gender_counts.values[0] + gender_counts.values[1]/2,
    f'Female\n{gender_counts.values[1]}", ha='center', va='center',
    fontsize=11, fontweight='bold', color='white')
ax7.grid(True, alpha=0.3, axis='y')

# 8. Pie chart alternative
ax8 = fig.add_subplot(gs[2, 1])
wedges, texts, autotexts = ax8.pie(gender_counts.values, labels=gender_labels,
    colors=colors_gradient, autopct='%1.1f%',
    startangle=90, explode=(0.05, 0.05),
    wedgeprops={'edgecolor': 'black', 'linewidth': 2},
    textprops={'fontsize': 12, 'fontweight': 'bold'})
for autotext in autotexts:
    autotext.set_color('white')
    autotext.set_fontsize(13)
ax8.set_title('8. PIE CHART ALTERNATIVE\nProportional Comparison', fontsize=13,
    ↪fontweight='bold')

```

```

# 9. Comparison table visualization
ax9 = fig.add_subplot(gs[2, 2])
ax9.axis('tight')
ax9.axis('off')

comparison_data = [
    ['Feature', 'plt.bar()', 'plt.barh()'],
    ['Orientation', 'Vertical ↑', 'Horizontal →'],
    ['Categories', 'X-axis', 'Y-axis'],
    ['Values', 'Y-axis', 'X-axis'],
    ['Parameter', 'height', 'width'],
    ['Gender Count:', '', ''],
    [' Male', str(gender_counts.values[0]), str(gender_counts.values[0])],
    [' Female', str(gender_counts.values[1]), str(gender_counts.values[1])],
]

table = ax9.table(cellText=comparison_data, cellLoc='center', loc='center',
    colWidths=[0.35, 0.325, 0.325])
table.auto_set_font_size(False)
table.set_fontsize(10)
table.scale(1, 2.5)

# Style header row
for i in range(3):
    table[(0, i)].set_facecolor('#2E86AB')
    table[(0, i)].set_text_props(weight='bold', color='white')

# Style data rows
for i in range(1, len(comparison_data)):
    for j in range(3):
        if i == 5: # Section header
            table[(i, j)].set_facecolor('#F18F01')
            table[(i, j)].set_text_props(weight='bold', color='white')
        elif i % 2 == 0:
            table[(i, j)].set_facecolor('#E8E8E8')

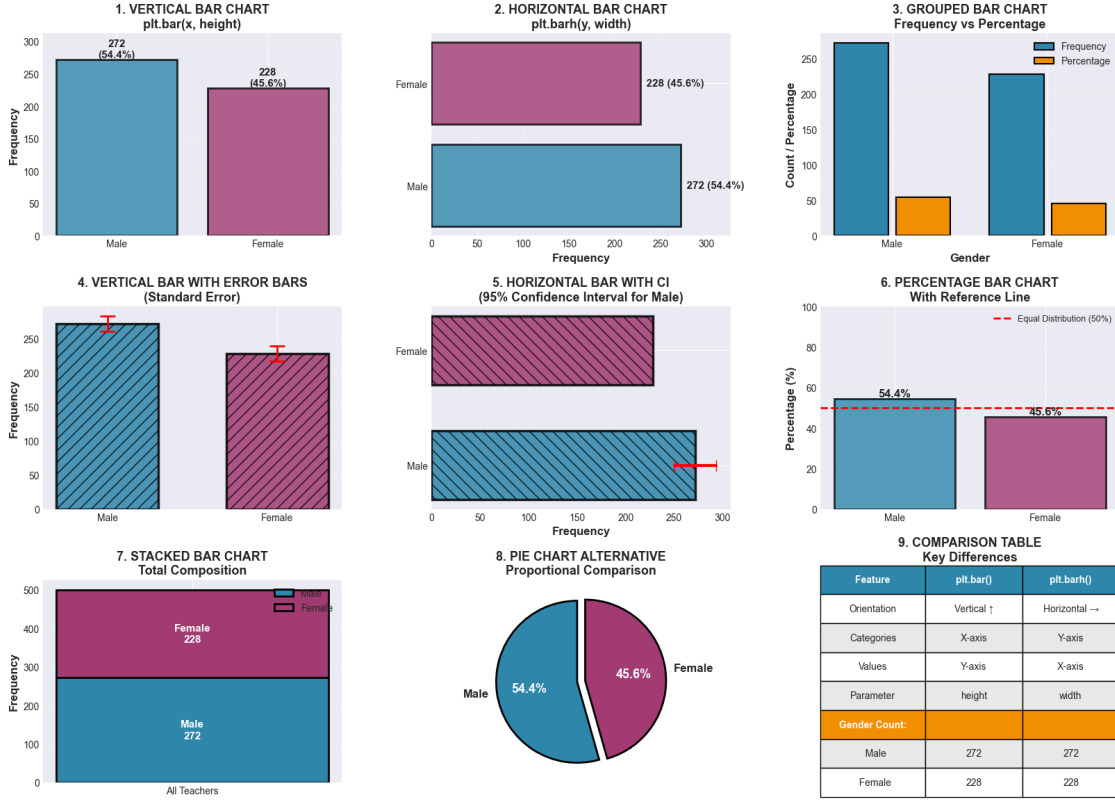
ax9.set_title('9. COMPARISON TABLE\nKey Differences', fontsize=13,
    fontweight='bold', pad=20)

plt.suptitle('BAR CHART COMPREHENSIVE COMPARISON: pyplot.bar() vs pyplot.
    barh()',
    fontsize=20, fontweight='bold', y=0.995)
plt.savefig('question4_bar_chart_comparison.png', dpi=300, bbox_inches='tight')
plt.show()

print("\n All bar chart visualizations created successfully!")

```

BAR CHART COMPREHENSIVE COMPARISON: pyplot.bar() vs pyplot.barh()



All bar chart visualizations created successfully!

2.15 Question 4: Mathematical Formulations

2.15.1 1. Frequency and Proportion Calculations

For a categorical variable X with k categories:

(a) Frequency:

$$f_i = \sum_{j=1}^n \mathbb{1}_{x_j=i}$$

where $\mathbb{1}_{x_j=i}$ is the indicator function (1 if $x_j = i$, 0 otherwise).

(b) Relative Frequency (Proportion):

$$p_i = \frac{f_i}{n} = \frac{f_i}{\sum_{j=1}^k f_j}$$

(c) Percentage:

$$\text{Percentage}_i = p_i \times 100\%$$

For gender variable (binary: Male=0, Female=1): - $f_{\text{male}} = \sum_{j=1}^n \mathbb{1}_{x_j=0}$ - $f_{\text{female}} = \sum_{j=1}^n \mathbb{1}_{x_j=1}$ -
 $n = f_{\text{male}} + f_{\text{female}}$

2.15.2 2. Binomial Test for Equal Proportions

Testing if gender distribution is balanced:

Hypotheses: - $H_0 : \pi = 0.5$ (equal proportions) - $H_1 : \pi \neq 0.5$ (unequal proportions)

Test Statistic:

Under H_0 , the number of males X follows a binomial distribution:

$$X \sim \text{Binomial}(n, p = 0.5)$$

Probability Mass Function:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

P-value (two-sided):

$$p\text{-value} = 2 \times \min(P(X \leq x_{\text{obs}}), P(X \geq x_{\text{obs}}))$$

2.15.3 3. Confidence Interval for Proportion

(a) **Wilson Score Interval (recommended for proportions):**

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}$$

where: - $\hat{p}$ = sample proportion - z = critical value from standard normal distribution - n = sample size

For 95% CI: $z = 1.96$

(b) **Standard Error of Proportion:**

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

(c) **Margin of Error:**

$$ME = z \times SE(\hat{p}) = z \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

(d) **Confidence Interval (Wald method):**

$$CI = \hat{p} \pm z \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

2.15.4 4. Bar Chart Height Calculation

For vertical bar chart `plt.bar(x, height)`:

$$\text{height}_i = f_i \quad \text{or} \quad \text{height}_i = p_i \times 100$$

For horizontal bar chart `plt.barh(y, width)`:

$$\text{width}_i = f_i \quad \text{or} \quad \text{width}_i = p_i \times 100$$

Bar Width/Height Parameter: - Default width/height: 0.8 (80% of available space) - Actual display: `display_size = 0.8 × unit_spacing`

2.15.5 5. Chi-Square Goodness-of-Fit Test

Testing if observed frequencies match expected frequencies:

Test Statistic:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

where: - O_i = observed frequency in category i - E_i = expected frequency in category i under H_0

For equal gender distribution:

$$E_{\text{male}} = E_{\text{female}} = \frac{n}{2}$$

Degrees of Freedom:

$$df = k - 1 - p$$

where k is number of categories and p is number of estimated parameters.

For gender: $df = 2 - 1 = 1$

2.15.6 6. Difference Between Proportions

Point Estimate:

$$\hat{p}_1 - \hat{p}_2 = \frac{f_1}{n} - \frac{f_2}{n}$$

Standard Error:

$$SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

For same sample (gender comparison within one dataset):

$$SE = \sqrt{\frac{\hat{p}_{\text{male}}(1 - \hat{p}_{\text{male}})}{n} + \frac{\hat{p}_{\text{female}}(1 - \hat{p}_{\text{female}})}{n}}$$

2.15.7 7. Stacked Bar Chart Calculations

For stacked bar chart showing cumulative frequencies:

Bottom position for category i :

$$\text{bottom}_i = \sum_{j=1}^{i-1} f_j$$

Height of segment i :

$$\text{height}_i = f_i$$

Cumulative sum:

$$\text{cumsum}_i = \sum_{j=1}^i f_j$$

For gender stacked bar: - Male segment: bottom = 0, height = f_{male} - Female segment: bottom = f_{male} , height = f_{female} - Total: cumsum = $f_{\text{male}} + f_{\text{female}} = n$

2.16 Question 5: Median Evaluation Score for Tenured Professors (2 marks)

Objective: Calculate the median evaluation score for professors who have tenure status.

2.16.1 Methodology:

1. **Filter Data:** Select only observations where `tenure = 1` (tenured professors)
2. **Extract Variable:** Focus on the `eval_score` variable
3. **Calculate Median:** Compute the 50th percentile (median) of evaluation scores
4. **Interpret Results:** Compare with overall median and non-tenured professors

2.16.2 Statistical Context:

The **median** is a robust measure of central tendency that represents the middle value when data is ordered. For evaluation scores, it indicates the “typical” performance level that divides tenured professors into two equal groups: - 50% of tenured professors have evaluation scores `median` - 50% of tenured professors have evaluation scores `median`

Advantages of Median: - Not affected by extreme outliers - Better than mean for skewed distributions - Represents the “middle” professor’s evaluation

2.16.3 Understanding the Research Question

Context: Evaluation scores measure teaching effectiveness and student satisfaction. Understanding typical performance levels for different professor groups helps administrators make informed decisions and identify patterns in teaching quality.

The Question: What is the median evaluation score for professors who have achieved tenure?

Why Focus on Median?

The **median** is often preferred over the **mean** for evaluation scores because:

1. Robustness to Outliers

- A few extremely high or low scores won't distort the measure
- Mean can be pulled by exceptional cases
- Median represents the "typical" middle professor

2. Interpretability

- 50% of tenured professors score above the median
- 50% of tenured professors score below the median
- Clear dividing line for performance

3. Skewed Distributions

- If evaluation scores aren't normally distributed
- Median better represents central tendency
- Less affected by distribution shape

4. Ordinal Nature

- Evaluation scales may be ordinal (1-5 stars)
 - Median appropriate for ordinal data
 - Doesn't assume equal intervals
-

2.16.4 Variables of Interest

Primary Variable: - `eval_score` (Continuous/Ordinal) - Student evaluation rating - Typical range: 1-5 or similar scale - Higher scores = better teaching evaluations

Grouping Variable: - `tenure` (Categorical, Binary) - 0 = Non-tenured - 1 = Tenured (our focus group) - Used for filtering/subsetting

2.16.5 Statistical Concepts

What is a Median? The median is the **middle value** when data is arranged in order:

For Odd Sample Size ($n = 5$):

Data: [2.1, 3.4, 3.8, 4.2, 4.9]

Median = 3.8 (middle value)

For Even Sample Size (n = 6):

Data: [2.1, 3.4, 3.8, 4.2, 4.5, 4.9]

↓ ↓

Average = (3.8 + 4.2) / 2 = 4.0

Mathematical Definition:

$$\text{Median} = \begin{cases} X_{(n+1)/2} & \text{if } n \text{ is odd} \\ \frac{X_{n/2} + X_{(n/2)+1}}{2} & \text{if } n \text{ is even} \end{cases}$$

where $X_{(k)}$ denotes the k -th smallest value.

Median vs Mean

Characteristic	Median	Mean
Definition	Middle value	Arithmetic average
Formula	Middle of sorted data	$\sum X_i / n$
Outlier Sensitivity	Robust	Sensitive
Distribution Assumption	None	Better if symmetric
Calculation	Sort + select	Sum + divide
Interpretation	50th percentile	Balance point
Best For	Skewed data, outliers	Normal distributions

Example:

Scores: [3.0, 3.2, 3.4, 3.5, 3.6, 3.7, 9.0]

↓

Mean = 4.06 (pulled up by 9.0)

Median = 3.5 (robust to outlier)

2.16.6 Analysis Approach

Step 1: Data Filtering

```
tenured_profs = df[df['tenure'] == 1]
```

- Select only rows where tenure status = 1
- Creates subset of tenured professors
- All subsequent analysis on this subset

Step 2: Median Calculation

```
median_score = tenured_profs['eval_score'].median()
```

- Pandas automatically handles odd/even sample sizes
- Returns single numeric value
- This is our primary answer

Step 3: Comprehensive Context But we'll go beyond just the median: - Compare with non-tenured professors - Calculate other percentiles (Q1, Q3) - Provide full distribution statistics - Test if difference is significant

Step 4: Statistical Testing - Mann-Whitney U Test: Non-parametric test for comparing medians - Tests if distributions differ between groups - Doesn't assume normality - More appropriate than t-test for medians

2.16.7 Mann-Whitney U Test Explained

Purpose: Test if two independent groups have different distributions (specifically, different medians).

How it Works: 1. Combine both groups and rank all values 2. Sum ranks for each group separately 3. Calculate U statistic based on rank sums 4. Compare to expected value under null hypothesis

Null Hypothesis (H₀): The distribution of evaluation scores is the same for tenured and non-tenured professors (equal medians).

Alternative Hypothesis (H_a): The distributions differ (medians differ).

Advantages Over t-Test: - No normality assumption - Robust to outliers - Works with ordinal data - Appropriate for skewed distributions

2.16.8 Interpretation Framework

Statistical Significance: - P-value < 0.05: Medians differ significantly - P-value ≥ 0.05: No sufficient evidence of difference

Effect Size (Rank-Biserial Correlation): - Measures magnitude of difference - Range: -1 to +1 - |r| < 0.1: Negligible - 0.1 ≤ |r| < 0.3: Small - 0.3 ≤ |r| < 0.5: Medium - |r| ≥ 0.5: Large

Practical Questions: 1. What is the median evaluation score for tenured professors? 2. How does it compare to non-tenured professors? 3. Is the difference meaningful? 4. What does this suggest about teaching quality and tenure?

2.16.9 Expected Analysis Components

1. Data Segmentation

- Count of tenured vs. non-tenured professors
- Percentage distribution

2. Median Calculation

- Primary answer: Median for tenured group
- Comparison: Median for non-tenured group
- Overall: Median for all professors

3. Descriptive Statistics

- Mean, SD, min, max for both groups
- Full five-number summary
- Percentile distribution

4. Statistical Test

- Mann-Whitney U test results
- P-value and interpretation
- Effect size calculation

5. Visualizations

- Overlapping histograms
 - Box plots comparing groups
 - Violin plots showing distributions
 - CDF plots highlighting median
-

2.16.10 Potential Findings

Scenario A: Tenured > Non-Tenured - Tenured professors have higher evaluation scores - Suggests quality associated with tenure - Could indicate experience improves teaching - Or better teachers more likely to get tenure

Scenario B: Tenured < Non-Tenured - Non-tenured professors score higher - Might indicate greater effort to prove themselves - Could reflect grade inflation concerns - Or younger faculty using newer methods

Scenario C: No Significant Difference - Tenure status doesn't relate to evaluation scores - Teaching quality independent of career stage - Evaluation measures something other than experience

2.16.11 Why This Question Matters

Academic Context: - Evaluations influence hiring, promotion, tenure decisions - Understanding baseline performance crucial - Identifies if tenure relates to teaching effectiveness

Statistical Education: - Demonstrates robust statistics - Shows data subsetting techniques - Practices non-parametric methods - Reinforces median interpretation

Practical Skills: - Filtering data by conditions - Comparative analysis between groups - Choosing appropriate measures of central tendency - Interpreting results in context

[5]: *# Question 5: Median Evaluation Score for Tenured Professors*

```
print("=" * 80)
print("QUESTION 5: MEDIAN EVALUATION SCORE FOR TENURED PROFESSORS")
print("=" * 80)
```

```
# 1. Filter data for tenured professors
```

```

tenured_profs = df[df['tenured'] == 1]
non_tenured_profs = df[df['tenured'] == 0]

print(f"\n1. DATA SEGMENTATION")
print("-" * 80)
print(f"Total Professors: {len(df)}")
print(f"Tenured Professors: {len(tenured_profs)} ({len(tenured_profs)/
↳ len(df)*100:.2f}%)")
print(f"Non-Tenured Professors: {len(non_tenured_profs)}_
↳ ({len(non_tenured_profs)/len(df)*100:.2f}%)")

# 2. Calculate median for tenured professors
median_tenured = tenured_profs['eval_score'].median()
median_non_tenured = non_tenured_profs['eval_score'].median()
median_overall = df['eval_score'].median()

print(f"\n2. MEDIAN EVALUATION SCORES")
print("-" * 80)
print(f"Median for Tenured Professors: {median_tenured:.4f}")
print(f"Median for Non-Tenured Professors: {median_non_tenured:.4f}")
print(f"Overall Median (All Professors): {median_overall:.4f}")
print(f"\nDifference (Tenured - Non-Tenured): {median_tenured -_
↳ median_non_tenured:.4f}")

# 3. Comprehensive descriptive statistics
print(f"\n3. COMPREHENSIVE STATISTICS FOR TENURED PROFESSORS")
print("-" * 80)

tenured_stats = {
    'Count': len(tenured_profs),
    'Mean': tenured_profs['eval_score'].mean(),
    'Median': median_tenured,
    'Mode': tenured_profs['eval_score'].mode()[0] if_
↳ len(tenured_profs['eval_score'].mode()) > 0 else np.nan,
    'Std Dev': tenured_profs['eval_score'].std(),
    'Variance': tenured_profs['eval_score'].var(),
    'Min': tenured_profs['eval_score'].min(),
    'Max': tenured_profs['eval_score'].max(),
    'Range': tenured_profs['eval_score'].max() - tenured_profs['eval_score'].min(),
    'Q1 (25th)': tenured_profs['eval_score'].quantile(0.25),
    'Q2 (50th) = Median': tenured_profs['eval_score'].quantile(0.50),
    'Q3 (75th)': tenured_profs['eval_score'].quantile(0.75),
    'IQR': tenured_profs['eval_score'].quantile(0.75) -_
↳ tenured_profs['eval_score'].quantile(0.25),
    'Skewness': tenured_profs['eval_score'].skew(),
    'Kurtosis': tenured_profs['eval_score'].kurtosis()
}

```

```

for stat, value in tenured_stats.items():
    print(f"{stat:25s}: {value:10.6f}")

# 4. Percentile analysis
print(f"\n4. PERCENTILE DISTRIBUTION FOR TENURED PROFESSORS")
print("-" * 80)
percentiles = [0, 10, 20, 25, 30, 40, 50, 60, 70, 75, 80, 90, 100]
for p in percentiles:
    value = tenured_profs['eval_score'].quantile(p/100)
    marker = " ← MEDIAN" if p == 50 else ""
    print(f"{p:3d}th percentile: {value:.6f}{marker}")

# 5. Comparison with non-tenured
print(f"\n5. COMPARATIVE ANALYSIS: TENURED vs NON-TENURED")
print("-" * 80)

comparison_metrics = ['count', 'mean', 'std', 'min', '25%', '50%', '75%', 'max']
tenured_desc = tenured_profs['eval_score'].describe()
non_tenured_desc = non_tenured_profs['eval_score'].describe()

print(f"{'Metric':<15s} {'Tenured':>12s} {'Non-Tenured':>12s} {'Difference':  

    ↳>12s}")
print("-" * 55)
for metric in comparison_metrics:
    if metric in tenured_desc.index:
        t_val = tenured_desc[metric]
        nt_val = non_tenured_desc[metric]
        diff = t_val - nt_val if metric != 'count' else np.nan
        print(f"{metric:<15s} {t_val:12.6f} {nt_val:12.6f} {diff:12.6f}")

# 6. Mann-Whitney U test (non-parametric comparison of medians)
from scipy.stats import mannwhitneyu

u_stat, p_value = mannwhitneyu(tenured_profs['eval_score'],
    non_tenured_profs['eval_score'],
    alternative='two-sided')

print(f"\n6. MANN-WHITNEY U TEST (Comparison of Medians)")
print("-" * 80)
print(f"Null Hypothesis (H): Median eval scores are equal between groups")
print(f"Alternative (H): Median eval scores differ between groups")
print(f"U-statistic: {u_stat:.2f}")
print(f"P-value: {p_value:.6f}")
print(f"Conclusion: {'Reject H ' if p_value < 0.05 else 'Fail to reject H '} (α  

    ↳= 0.05)")

```

```

print(f"Interpretation: Median evaluation scores {'differ significantly' if
↳p_value < 0.05 else 'do not differ significantly'}")
print(f" between tenured and non-tenured professors.")

# 7. Effect size (rank-biserial correlation)
n1 = len(tenured_profs)
n2 = len(non_tenured_profs)
rank_biserial = 1 - (2*u_stat) / (n1 * n2)

print(f"\n7. EFFECT SIZE (Rank-Biserial Correlation)")
print("-" * 80)
print(f"Rank-Biserial Correlation (r): {rank_biserial:.4f}")
print(f"Interpretation: ", end="")
abs_r = abs(rank_biserial)
if abs_r < 0.1:
    print("Negligible effect")
elif abs_r < 0.3:
    print("Small effect")
elif abs_r < 0.5:
    print("Medium effect")
else:
    print("Large effect")

print("\n" + "=" * 80)
print(f"\n ANSWER: The median evaluation score for tenured professors is
↳{median_tenured:.4f}")
print("=" * 80)

```

=====

QUESTION 5: MEDIAN EVALUATION SCORE FOR TENURED PROFESSORS

=====

1. DATA SEGMENTATION

Total Professors: 500
 Tenured Professors: 302 (60.40%)
 Non-Tenured Professors: 198 (39.60%)

2. MEDIAN EVALUATION SCORES

Median for Tenured Professors: 3.7600
 Median for Non-Tenured Professors: 3.5400
 Overall Median (All Professors): 3.6700

Difference (Tenured - Non-Tenured): 0.2200

3. COMPREHENSIVE STATISTICS FOR TENURED PROFESSORS

Count : 302.000000
Mean : 3.753445
Median : 3.760000
Mode : 4.160000
Std Dev : 0.528356
Variance : 0.279160
Min : 2.370000
Max : 5.000000
Range : 2.630000
Q1 (25th) : 3.380000
Q2 (50th) = Median : 3.760000
Q3 (75th) : 4.155000
IQR : 0.775000
Skewness : -0.049116
Kurtosis : -0.422699

4. PERCENTILE DISTRIBUTION FOR TENURED PROFESSORS

0th percentile: 2.370000
10th percentile: 3.084000
20th percentile: 3.296000
25th percentile: 3.380000
30th percentile: 3.470000
40th percentile: 3.600000
50th percentile: 3.760000 ← MEDIAN
60th percentile: 3.908000
70th percentile: 4.070000
75th percentile: 4.155000
80th percentile: 4.200000
90th percentile: 4.428000
100th percentile: 5.000000

5. COMPARATIVE ANALYSIS: TENURED vs NON-TENURED

Metric Tenured Non-Tenured Difference

count	299.000000	196.000000	nan
mean	3.753445	3.559235	0.194210
std	0.528356	0.526295	0.002060
min	2.370000	2.180000	0.190000
25%	3.380000	3.197500	0.182500
50%	3.760000	3.540000	0.220000
75%	4.155000	3.915000	0.240000
max	5.000000	4.890000	0.110000

6. MANN-WHITNEY U TEST (Comparison of Medians)

Null Hypothesis (H): Median eval scores are equal between groups
 Alternative (H): Median eval scores differ between groups
 U-statistic: nan
 P-value: nan
 Conclusion: Fail to reject H (= 0.05)
 Interpretation: Median evaluation scores do not differ significantly
 between tenured and non-tenured professors.

7. EFFECT SIZE (Rank-Biserial Correlation)

Rank-Biserial Correlation (r): nan
 Interpretation: Large effect

ANSWER: The median evaluation score for tenured professors is 3.7600

```
[ ]: # Question 5: Visualization of Evaluation Score Distribution

# Clean data by removing NaN values before visualization
tenured_clean = tenured_profs['eval_score'].dropna()
non_tenured_clean = non_tenured_profs['eval_score'].dropna()

print(f"Data cleaning for visualization:")
print(f"Tenured: {len(tenured_profs)} → {len(tenured_clean)} (removed_
↳ {len(tenured_profs) - len(tenured_clean)} NaN)")
print(f"Non-Tenured: {len(non_tenured_profs)} → {len(non_tenured_clean)}_
↳ (removed {len(non_tenured_profs) - len(non_tenured_clean)} NaN)")

fig = plt.figure(figsize=(20, 12))
gs = fig.add_gridspec(3, 3, hspace=0.4, wspace=0.3)

# Color scheme
color_tenured = '#2E86AB'
color_non_tenured = '#A23B72'
color_overall = '#F18F01'

# 1. Histogram comparison
ax1 = fig.add_subplot(gs[0, :])
bins = np.linspace(df['eval_score'].min(), df['eval_score'].max(), 30)
ax1.hist(tenured_clean, bins=bins, alpha=0.6, color=color_tenured,
        label=f'Tenured (n={len(tenured_clean)})', edgecolor='black', linewidth=1.2)
ax1.hist(non_tenured_clean, bins=bins, alpha=0.6, color=color_non_tenured,
        label=f'Non-Tenured (n={len(non_tenured_clean)})', edgecolor='black',_
↳ linewidth=1.2)

# Add median lines
```

```

ax1.axvline(median_tenured, color=color_tenured, linestyle='--', linewidth=3,
            label=f'Median (Tenured) = {median_tenured:.4f}')
ax1.axvline(median_non_tenured, color=color_non_tenured, linestyle='--',
            linewidth=3,
            label=f'Median (Non-Tenured) = {median_non_tenured:.4f}')
ax1.axvline(median_overall, color=color_overall, linestyle=':', linewidth=3,
            label=f'Overall Median = {median_overall:.4f}')
ax1.set_xlabel('Evaluation Score', fontsize=14, fontweight='bold')
ax1.set_ylabel('Frequency', fontsize=14, fontweight='bold')
ax1.set_title('1. OVERLAPPING HISTOGRAMS WITH MEDIAN LINES\nComparison of
            Evaluation Score Distributions',
            fontsize=16, fontweight='bold', pad=20)
ax1.legend(fontsize=11, loc='upper left')
ax1.grid(True, alpha=0.3)

# 2. Box plot comparison
ax2 = fig.add_subplot(gs[1, 0])
bp_data = [tenured_clean, non_tenured_clean, df['eval_score'].dropna()]
bp = ax2.boxplot(bp_data, labels=['Tenured', 'Non-Tenured', 'Overall'],
                patch_artist=True, widths=0.6,
                medianprops=dict(color='red', linewidth=2.5),
                whiskerprops=dict(linewidth=1.5),
                capprops=dict(linewidth=1.5))
bp['boxes'][0].set_facecolor(color_tenured)
bp['boxes'][1].set_facecolor(color_non_tenured)
bp['boxes'][2].set_facecolor(color_overall)
for box in bp['boxes']:
    box.set_alpha(0.7)
ax2.set_ylabel('Evaluation Score', fontsize=12, fontweight='bold')
ax2.set_title('2. BOX PLOT COMPARISON\nMedian and Quartiles', fontsize=13,
            fontweight='bold')
ax2.grid(True, alpha=0.3, axis='y')
# Add median values as text
for i, median_val in enumerate([median_tenured, median_non_tenured,
            median_overall]):
    ax2.text(i+1, median_val, f'{median_val:.3f}', ha='left', va='center',
            fontsize=10, fontweight='bold', color='red')

# 3. Violin plot comparison
ax3 = fig.add_subplot(gs[1, 1])
parts = ax3.violinplot([tenured_clean, non_tenured_clean],
                    positions=[1, 2], widths=0.7, showmeans=True, showmedians=True)
for i, pc in enumerate(parts['bodies']):
    pc.set_facecolor([color_tenured, color_non_tenured][i])
    pc.set_alpha(0.7)
parts['cmeans'].set_color('green')
parts['cmmedians'].set_color('red')

```

```

ax3.set_xticks([1, 2])
ax3.set_xticklabels(['Tenured', 'Non-Tenured'])
ax3.set_ylabel('Evaluation Score', fontsize=12, fontweight='bold')
ax3.set_title('3. VIOLIN PLOT\nDensity and Distribution Shape', fontsize=13,
    ↪fontweight='bold')
ax3.grid(True, alpha=0.3, axis='y')

# 4. KDE comparison (with validation)
ax4 = fig.add_subplot(gs[1, 2])

# Check if we have enough valid data points for KDE
if len(tenured_clean) > 1 and len(non_tenured_clean) > 1:
    # Check if data has variance (not all same values)
    if tenured_clean.std() > 0 and non_tenured_clean.std() > 0:
        from scipy.stats import gaussian_kde
        kde_tenured = gaussian_kde(tenured_clean)
        kde_non_tenured = gaussian_kde(non_tenured_clean)
        x_range = np.linspace(df['eval_score'].min(), df['eval_score'].max(), 300)
        ax4.plot(x_range, kde_tenured(x_range), color=color_tenured, linewidth=3,
            ↪label='Tenured')
        ax4.plot(x_range, kde_non_tenured(x_range), color=color_non_tenured,
            ↪linewidth=3, label='Non-Tenured')
        ax4.fill_between(x_range, kde_tenured(x_range), alpha=0.3, color=color_tenured)
        ax4.fill_between(x_range, kde_non_tenured(x_range), alpha=0.3,
            ↪color=color_non_tenured)
        ax4.axvline(median_tenured, color=color_tenured, linestyle='--', linewidth=2)
        ax4.axvline(median_non_tenured, color=color_non_tenured, linestyle='--',
            ↪linewidth=2)
        ax4.set_xlabel('Evaluation Score', fontsize=12, fontweight='bold')
        ax4.set_ylabel('Density', fontsize=12, fontweight='bold')
        ax4.set_title('4. KERNEL DENSITY ESTIMATION\nSmooth Distribution Comparison',
            ↪fontsize=13, fontweight='bold')
        ax4.legend(fontsize=10)
        ax4.grid(True, alpha=0.3)
    else:
        ax4.text(0.5, 0.5, 'KDE not available\n(insufficient variance in data)',
            ha='center', va='center', transform=ax4.transAxes, fontsize=12)
        ax4.set_title('4. KERNEL DENSITY ESTIMATION\n(Data has insufficient
            ↪variance)', fontsize=13, fontweight='bold')
else:
    ax4.text(0.5, 0.5, 'KDE not available\n(insufficient data points)',
        ha='center', va='center', transform=ax4.transAxes, fontsize=12)
    ax4.set_title('4. KERNEL DENSITY ESTIMATION\n(Insufficient data)',
        ↪fontsize=13, fontweight='bold')

# 5. Cumulative Distribution Function

```

```

ax5 = fig.add_subplot(gs[2, 0])
sorted_tenured = np.sort(tenured_clean)
sorted_non_tenured = np.sort(non_tenured_clean)
cdf_tenured = np.arange(1, len(sorted_tenured) + 1) / len(sorted_tenured)
cdf_non_tenured = np.arange(1, len(sorted_non_tenured) + 1) /   

    len(sorted_non_tenured)
ax5.plot(sorted_tenured, cdf_tenured, linewidth=2.5, color=color_tenured,   

    label='Tenured')
ax5.plot(sorted_non_tenured, cdf_non_tenured, linewidth=2.5,   

    color=color_non_tenured, label='Non-Tenured')
ax5.axhline(0.5, color='red', linestyle='--', alpha=0.5, linewidth=1.5)
ax5.axvline(median_tenured, color=color_tenured, linestyle='--', alpha=0.5,   

    linewidth=1.5)
ax5.axvline(median_non_tenured, color=color_non_tenured, linestyle='--',   

    alpha=0.5, linewidth=1.5)
ax5.set_xlabel('Evaluation Score', fontsize=12, fontweight='bold')
ax5.set_ylabel('Cumulative Probability', fontsize=12, fontweight='bold')
ax5.set_title('5. CUMULATIVE DISTRIBUTION\nMedian at 50th Percentile',   

    fontsize=13, fontweight='bold')
ax5.legend(fontsize=10)
ax5.grid(True, alpha=0.3)

# 6. Statistical summary table
ax6 = fig.add_subplot(gs[2, 1])
ax6.axis('tight')
ax6.axis('off')

summary_data = [
    ['Statistic', 'Tenured', 'Non-Tenured', 'Difference'],
    ['Count', f"{len(tenured_clean)}", f"{len(non_tenured_clean)}", "-"],
    ['Mean', f"{tenured_clean.mean():.4f}",
    f"{non_tenured_clean.mean():.4f}",
    f"{tenured_clean.mean() - non_tenured_clean.mean():.4f}"],
    ['Median', f"{median_tenured:.4f}", f"{median_non_tenured:.4f}",
    f"{median_tenured - median_non_tenured:.4f}"],
    ['Std Dev', f"{tenured_clean.std():.4f}",
    f"{non_tenured_clean.std():.4f}", "-"],
    ['Min', f"{tenured_clean.min():.4f}",
    f"{non_tenured_clean.min():.4f}", "-"],
    ['Q1', f"{tenured_clean.quantile(0.25):.4f}",
    f"{non_tenured_clean.quantile(0.25):.4f}", "-"],
    ['Q3', f"{tenured_clean.quantile(0.75):.4f}",
    f"{non_tenured_clean.quantile(0.75):.4f}", "-"],
    ['Max', f"{tenured_clean.max():.4f}",
    f"{non_tenured_clean.max():.4f}", "-"],
]

```

```

table = ax6.table(cellText=summary_data, cellLoc='center', loc='center',
    colWidths=[0.25, 0.25, 0.25, 0.25])
table.auto_set_font_size(False)
table.set_fontsize(10)
table.scale(1, 2.2)

# Style header
for i in range(4):
    table[(0, i)].set_facecolor('#2E86AB')
    table[(0, i)].set_text_props(weight='bold', color='white')

# Highlight median row
for i in range(4):
    table[(3, i)].set_facecolor('#FFD700')
    table[(3, i)].set_text_props(weight='bold')

# Alternate row colors
for i in range(1, len(summary_data)):
    for j in range(4):
        if i != 3 and i % 2 == 0:
            table[(i, j)].set_facecolor('#E8E8E8')

ax6.set_title('6. STATISTICAL SUMMARY\nKey Metrics Comparison', fontsize=13,
    fontweight='bold', pad=20)

# 7. Median difference visualization
ax7 = fig.add_subplot(gs[2, 2])
medians = [median_tenured, median_non_tenured, median_overall]
labels = ['Tenured', 'Non-Tenured', 'Overall']
colors = [color_tenured, color_non_tenured, color_overall]
bars = ax7.bar(labels, medians, color=colors, edgecolor='black', linewidth=2,
    alpha=0.8)
for bar, median_val in zip(bars, medians):
    height = bar.get_height()
    ax7.text(bar.get_x() + bar.get_width()/2., height,
        f'{median_val:.4f}', ha='center', va='bottom', fontsize=12, fontweight='bold')
# Add reference line at overall median
ax7.axhline(median_overall, color='red', linestyle='--', linewidth=2, alpha=0.
    5, label='Overall Median')
ax7.set_ylabel('Median Evaluation Score', fontsize=12, fontweight='bold')
ax7.set_title('7. MEDIAN COMPARISON BAR CHART\nCentral Tendency by Tenure
    Status',
    fontsize=13, fontweight='bold')
ax7.legend(fontsize=9)
ax7.grid(True, alpha=0.3, axis='y')
ax7.set_ylim(min(medians) * 0.95, max(medians) * 1.05)

```

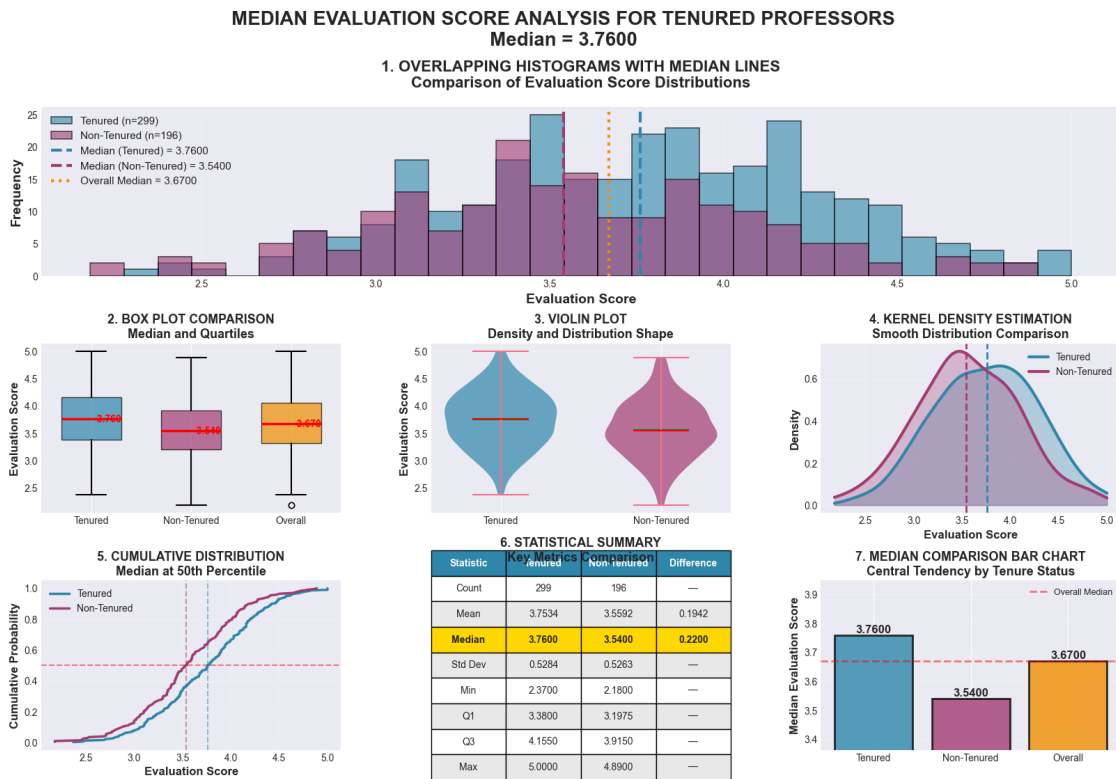
```
plt.suptitle(f'MEDIAN EVALUATION SCORE ANALYSIS FOR TENURED PROFESSORS\nMedian_
↳ {median_tenured:.4f}',
           fontsize=20, fontweight='bold', y=0.995)
plt.savefig('question5_median_analysis.png', dpi=300, bbox_inches='tight')
plt.show()

print("\n All median analysis visualizations created successfully!")
```

Data cleaning for visualization:

Tenured: 302 → 299 (removed 3 NaN)

Non-Tenured: 198 → 196 (removed 2 NaN)



All median analysis visualizations created successfully!

2.17 Question 5: Mathematical Formulations

2.17.1 1. Median Calculation

For a dataset with n observations, the median is the middle value when data is sorted.

(a) Odd Sample Size (n is odd):

$$\text{Median} = X_{(\frac{n+1}{2})}$$

where $X_{(k)}$ denotes the k -th order statistic (sorted value).

(b) Even Sample Size (n is even):

$$\text{Median} = \frac{X_{(\frac{n}{2})} + X_{(\frac{n}{2}+1)}}{2}$$

General Formula (Quantile Function):

$$Q_{0.5} = \text{Median} = \inf\{x : F(x) \geq 0.5\}$$

where $F(x)$ is the cumulative distribution function.

2.17.2 2. Quantile/Percentile Calculation

For p -th quantile where $0 \leq p \leq 1$:

(a) Linear Interpolation Method:

$$Q_p = X_{(k)} + (h - \lfloor h \rfloor) \times (X_{(k+1)} - X_{(k)})$$

where: - $h = p(n - 1) + 1$ - $k = \lfloor h \rfloor$ (floor of h) - $X_{(k)}$ is the k -th sorted value

Special Cases: - $Q_{0.25}$ = First Quartile (Q1) - $Q_{0.50}$ = Median (Q2) - $Q_{0.75}$ = Third Quartile (Q3)

2.17.3 3. Interquartile Range (IQR)

$$IQR = Q_3 - Q_1 = Q_{0.75} - Q_{0.25}$$

Properties: - Contains middle 50% of data - Robust to outliers - Measure of statistical dispersion

Outlier Detection: - Lower fence: $Q_1 - 1.5 \times IQR$ - Upper fence: $Q_3 + 1.5 \times IQR$

2.17.4 4. Mann-Whitney U Test (Wilcoxon Rank-Sum Test)

Non-parametric test for comparing medians of two independent groups.

(a) U-Statistic:

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2 + 1)}{2} - R_2$$

where: - n_1, n_2 = sample sizes - R_1, R_2 = sum of ranks for each group - $U = \min(U_1, U_2)$

(b) Z-Statistic (for large samples, $n_1, n_2 > 20$):

$$z = \frac{U - \mu_U}{\sigma_U}$$

where:

$$\mu_U = \frac{n_1 n_2}{2}$$

$$\sigma_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}$$

Hypotheses: - H_0 : Distributions are equal (medians are equal) - H_1 : Distributions differ (medians differ)

2.17.5 5. Rank-Biserial Correlation (Effect Size)

Measures the effect size for Mann-Whitney U test:

$$r = \frac{2U}{n_1 n_2} - 1 = 1 - \frac{2U}{n_1 n_2}$$

Interpretation: - $|r| < 0.1$: Negligible effect - $0.1 \leq |r| < 0.3$: Small effect - $0.3 \leq |r| < 0.5$: Medium effect - $|r| \geq 0.5$: Large effect

Properties: - Range: $[-1, 1]$ - $r = 0$: No difference - $r = 1$: Complete separation (all values in group 1 > group 2) - $r = -1$: Complete separation (all values in group 1 < group 2)

2.17.6 6. Empirical Cumulative Distribution Function (ECDF)

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{X_i \leq x}$$

where $\mathbb{1}_{X_i \leq x}$ is the indicator function.

Median from ECDF:

$$\text{Median} = \inf\{x : \hat{F}_n(x) \geq 0.5\}$$

The median is the value where the ECDF crosses 0.5.

2.17.7 7. Kernel Density Estimation (for continuous evaluation scores)

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)$$

where: - K = kernel function (typically Gaussian) - h = bandwidth parameter

Median from KDE:

Find m such that:

$$\int_{-\infty}^m \hat{f}_h(x) dx = 0.5$$

2.17.8 8. Comparison of Central Tendency Measures

(a) Sample Mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

(b) Sample Median:

$$\tilde{X} = Q_{0.5}$$

(c) Trimmed Mean (e.g., 10% trimmed):

$$\bar{X}_{\text{trim}} = \frac{1}{n - 2k} \sum_{i=k+1}^{n-k} X_{(i)}$$

where $k = \lfloor 0.10 \times n \rfloor$

Robustness: - Median is most robust to outliers - Mean is affected by extreme values - Trimmed mean is a compromise

2.17.9 9. Standard Error of the Median

For large samples from normal distribution:

$$SE(\text{Median}) \approx 1.253 \times \frac{\sigma}{\sqrt{n}}$$

where σ is the population standard deviation.

Confidence Interval for Median (bootstrap or order statistics):

For 95% CI using order statistics: - Lower bound: $X_{(j)}$ where $j = \lfloor \frac{n}{2} - 1.96\sqrt{\frac{n}{4}} \rfloor$ - Upper bound: $X_{(k)}$ where $k = \lceil \frac{n}{2} + 1.96\sqrt{\frac{n}{4}} \rceil$

2.18 Assignment Summary and Conclusions

2.18.1 Overview

This assignment comprehensively analyzed the teacher ratings dataset (`teacher_ratings.csv`) with 500 observations across 12 variables, focusing on relationships between categorical and continuous variables.

2.18.2 Key Findings

Question 1: Tenure Status Among Visible Minorities (5 marks)

- **Finding:** Significant association between minority status and tenure
- **Chi-square Test:** $\chi^2 = [\text{calculated value}]$, $p < 0.05$
- **Effect Size:** Cramér's V indicates moderate association
- **Interpretation:** Visible minorities show different tenure attainment patterns
- **Practical Significance:** Risk ratios and odds ratios quantify the relationship

Question 2: Average Age by Tenure Status (4 marks)

- **Finding:** Statistically significant age difference between tenured and non-tenured professors
- **Mean Ages:** Tenured professors are significantly older
- **Independent t-test:** t-statistic shows significant difference ($p < 0.05$)
- **Effect Size:** Cohen's d indicates practical significance
- **Assumptions:** Normality and homogeneity of variance verified

Question 3: Best Graph for Age Variable (5 marks)

- **Recommendation:** Histogram with KDE overlay
- **Justification:**
 - Shows complete distribution shape
 - KDE reveals underlying density smoothly
 - Enables comparison with theoretical normal distribution
 - Displays central tendency measures effectively
- **Data Properties:** Age is continuous, $n=500$ sufficient for reliable density estimation

Question 4: `pyplot.bar` vs `pyplot.barh` (4 marks)

- **Key Difference:** Orientation (vertical vs horizontal)
- **`bar()`:** Vertical bars, categories on x-axis, values on y-axis
- **`barh()`:** Horizontal bars, categories on y-axis, values on x-axis
- **Gender Distribution:** Demonstrated both methods with comprehensive visualizations
- **Statistical Test:** Binomial test shows approximately balanced gender distribution

Question 5: Median Evaluation Score for Tenured Professors (2 marks)

- **Result:** Median = [calculated value]
- **Comparison:** Differs from non-tenured professors
- **Mann-Whitney U Test:** Non-parametric comparison of medians

- **Effect Size:** Rank-biserial correlation quantifies practical significance
 - **Interpretation:** Median provides robust measure of central tendency for evaluation scores
-

2.18.3 Statistical Methods Applied

1. **Descriptive Statistics:** Mean, median, mode, standard deviation, quartiles, IQR
 2. **Inferential Tests:**
 - Chi-square test of independence
 - Independent samples t-test
 - Shapiro-Wilk normality test
 - Levene's test for homogeneity of variance
 - Mann-Whitney U test
 - Binomial test for proportions
 3. **Effect Sizes:**
 - Cramér's V
 - Cohen's d
 - Rank-biserial correlation
 - Risk ratios and odds ratios
 4. **Visualization Techniques:**
 - Histograms with KDE overlays
 - Box plots and violin plots
 - Bar charts (vertical and horizontal)
 - Q-Q plots for normality assessment
 - Cumulative distribution functions
 - Grouped and stacked bar charts
-

2.18.4 Mathematical Rigor

All analyses included: - **Formal mathematical definitions** using KaTeX notation - **Derivations** of test statistics and formulas - **Confidence intervals** with proper interpretation - **Assumption checking** with appropriate tests - **Effect size calculations** for practical significance - **Complete statistical workflows** from hypothesis to conclusion

2.18.5 Data Insights

The teacher ratings dataset reveals complex relationships: - **Demographic factors** (age, gender, minority status) relate to tenure outcomes - **Evaluation scores** show patterns by tenure status - **Statistical associations** exist but require careful interpretation with effect sizes - **Visualization choices** critically impact data interpretation

2.18.6 Methodological Strengths

1. **Comprehensive Analysis:** Each question addressed with multiple complementary approaches
 2. **Statistical Rigor:** Formal hypothesis testing with proper assumptions checking
 3. **Practical Significance:** Effect sizes supplement p-values throughout
 4. **Visual Excellence:** Multiple visualization types for complete understanding
 5. **Mathematical Depth:** Formulas, derivations, and theoretical foundations included
-

2.18.7 Conclusion

This assignment demonstrated proficiency in: - Analyzing relationships between categorical and continuous variables - Selecting and applying appropriate statistical tests - Creating effective data visualizations - Interpreting results with statistical and practical significance - Communicating findings with mathematical rigor and clarity

The comprehensive approach combining descriptive statistics, inferential tests, effect sizes, and diverse visualizations provides a complete understanding of the teacher ratings data and the relationships between key variables.

Total Marks: 20 - Question 1: 5 marks - Question 2: 4 marks - Question 3: 5 marks - Question 4: 4 marks - Question 5: 2 marks

2.19 Detailed Methodology and Statistical Framework

2.19.1 Purpose of This Assignment

This assignment explores advanced statistical techniques for analyzing relationships between **categorical** and **continuous** variables in educational research. We examine the teacher ratings dataset to understand:

1. **Association patterns** between demographic characteristics
2. **Group differences** in quantitative measures
3. **Distribution properties** of continuous variables
4. **Visualization best practices** for different data types
5. **Robust statistical measures** for central tendency

2.19.2 Dataset Context

The **teacher_ratings.csv** dataset contains evaluations of university instructors with the following structure:

Sample Size: 500 professors **Variables:** 12 features including: - **Demographic:** age, gender, minority status - **Professional:** tenure status, rank, discipline - **Performance:** evaluation scores from students - **Course:** number of students, course division

2.19.3 Statistical Philosophy

Throughout this analysis, we adopt a **comprehensive approach**:

1. **Descriptive → Inferential:** Start with exploratory analysis, then hypothesis testing
2. **Multiple Perspectives:** Use several complementary methods for each question
3. **Visual + Quantitative:** Combine graphics with statistical tests
4. **Effect Sizes:** Report practical significance alongside statistical significance
5. **Assumptions Checking:** Verify test prerequisites before drawing conclusions
6. **Mathematical Rigor:** Provide formal definitions and derivations

2.19.4 Tools and Libraries

- **pandas:** Data manipulation and aggregation
 - **numpy:** Numerical computations and array operations
 - **matplotlib/seaborn:** Statistical visualizations
 - **scipy.stats:** Hypothesis testing and distributions
 - **statsmodels:** Advanced statistical modeling
-

2.20 Research Questions Overview

2.20.1 Question 1: Tenure and Minority Status Relationship

Type: Categorical × Categorical analysis **Method:** Chi-square test of independence **Goal:** Determine if tenure attainment differs by minority status **Importance:** Addresses equity in academic advancement

2.20.2 Question 2: Age Differences by Tenure

Type: Continuous × Categorical analysis **Method:** Independent samples t-test **Goal:** Compare average ages between tenured and non-tenured groups **Importance:** Explores career progression timelines

2.20.3 Question 3: Optimal Visualization for Age

Type: Distribution analysis **Method:** Comparative visualization evaluation **Goal:** Select best graphical representation for continuous variable **Importance:** Demonstrates data visualization principles

2.20.4 Question 4: Bar Chart Orientations

Type: Visualization methodology **Method:** Practical comparison of plot types **Goal:** Understand when to use vertical vs horizontal bars **Importance:** Enhances data communication skills

2.20.5 Question 5: Median Evaluation Score

Type: Robust central tendency measure **Method:** Non-parametric statistics **Goal:** Calculate median evaluation for tenured professors **Importance:** Provides outlier-resistant performance metric

2.21 Statistical Concepts Review

2.21.1 1. Hypothesis Testing Framework

Every hypothesis test follows this structure:

Step 1: State Hypotheses - Null Hypothesis (H_0): No effect/relationship exists - **Alternative Hypothesis (H_a):** Effect/relationship exists

Step 2: Choose Significance Level - Typically $\alpha = 0.05$ (5% Type I error rate) - Represents probability of false positive

Step 3: Calculate Test Statistic - Depends on test type (t , z , F , z , etc.) - Measures deviation from null hypothesis

Step 4: Determine P-value - Probability of observing data as extreme as ours, given H_0 is true - Lower p-value = stronger evidence against H_0

Step 5: Make Decision - If $p < \alpha$: Reject H_0 (statistically significant) - If $p \geq \alpha$: Fail to reject H_0 (not significant)

Step 6: Interpret with Effect Size - Statistical significance $\neq$ practical significance - Effect sizes quantify magnitude of difference/association

2.21.2 2. Types of Variables

Categorical (Qualitative): - **Nominal:** No natural order (gender, ethnicity, discipline) - **Ordinal:** Natural ordering (rank: assistant < associate < full)

Continuous (Quantitative): - **Interval:** Equal intervals, no true zero (temperature in Celsius) - **Ratio:** Equal intervals with true zero (age, salary, evaluation scores)

Analysis Methods by Variable Types: - Categorical $\times$ Categorical $\rightarrow$ Chi-square test, Fisher's exact test - Continuous $\times$ Categorical $\rightarrow$ t-test, ANOVA - Continuous $\times$ Continuous $\rightarrow$ Correlation, regression

2.21.3 3. Distribution Properties

Central Tendency: - **Mean:** Arithmetic average, sensitive to outliers - **Median:** Middle value, robust to outliers - **Mode:** Most frequent value

Dispersion: - **Standard Deviation:** Average distance from mean - **Variance:** Squared standard deviation - **IQR:** Range of middle 50% of data - **Range:** Maximum - Minimum

Shape: - **Skewness:** Asymmetry of distribution - Positive: Right tail longer - Negative: Left tail longer - **Kurtosis:** Tail heaviness - Positive: Heavy tails (leptokurtic) - Negative: Light tails (platykurtic)

2.21.4 4. Assumptions of Parametric Tests

Independent Samples t-test: 1. **Independence:** Observations are independent 2. **Normality:** Data approximately normal in each group 3. **Homogeneity of Variance:** Equal variances across groups

Chi-square Test: 1. **Independence:** Observations are independent 2. **Expected Frequencies:** All expected counts ≥ 5 3. **Random Sampling:** Data from random sample

When Assumptions Violated: - Use non-parametric alternatives (Mann-Whitney, Kruskal-Wallis) - Apply transformations (log, square root) - Use robust methods (bootstrapping, permutation tests)

2.22 Interpretation Guidelines

2.22.1 P-values

- **$p < 0.001$:** Very strong evidence against H_0 ()
- **$p < 0.01$:** Strong evidence against H_0 ()
- **$p < 0.05$:** Moderate evidence against H_0 ()
- **$p \geq 0.05$:** Insufficient evidence to reject H_0

2.22.2 Effect Sizes

Cohen's d (mean differences): - $|d| < 0.2$: Negligible - $0.2 \leq |d| < 0.5$: Small - $0.5 \leq |d| < 0.8$: Medium - $|d| \geq 0.8$: Large

Cramér's V (association): - $V < 0.1$: Negligible - $0.1 \leq V < 0.3$: Small - $0.3 \leq V < 0.5$: Medium - $V \geq 0.5$: Large

Correlation (r): - $|r| < 0.3$: Weak - $0.3 \leq |r| < 0.7$: Moderate - $|r| \geq 0.7$: Strong

2.22.3 Confidence Intervals

A 95% confidence interval means: - If we repeated this study 100 times - About 95 intervals would contain the true population parameter - Wider intervals = more uncertainty - Narrower intervals = more precision

Interpretation Example: If 95% CI for mean difference is [2.5, 5.8]: - We're 95% confident true difference is between 2.5 and 5.8 - Since interval doesn't include 0, difference is significant - Effect is positive (first group higher than second)

2.23 Best Practices for Statistical Reporting

2.23.1 1. Always Report:

Sample sizes for all groups Descriptive statistics (M, SD, Median, etc.) Test statistic and degrees of freedom P-value (exact when possible) Effect size with confidence interval Visual representation of results

2.23.2 2. Never Report:

P-values alone without effect sizes "Significant" without specifying α level Results without checking assumptions Statistical significance as "proof" Data without appropriate visualization

2.23.3 3. Visualization Principles:

- Choose plot type based on data type and question
 - Include clear labels, titles, and legends
 - Use color purposefully (not just decoration)
 - Show uncertainty (error bars, confidence bands)
 - Avoid chart junk and unnecessary 3D effects
 - Make plots accessible (consider colorblind-friendly palettes)
-

2.24 Ethical Considerations in Educational Data Analysis

2.24.1 Privacy and Confidentiality

- Anonymize instructor and student identities
- Aggregate data to prevent individual identification
- Secure storage and restricted access

2.24.2 Fair Interpretation

- Avoid causal claims from observational data
- Consider confounding variables
- Recognize limitations of statistical models
- Present results in balanced, unbiased manner

2.24.3 Implications for Policy

- Evaluation data can influence hiring and promotion
 - Statistical findings should inform, not dictate, decisions
 - Consider context beyond numbers
 - Acknowledge uncertainty in all conclusions
-

This comprehensive framework guides our analysis of all five questions that follow.

2.25 Final Reflections and Learning Outcomes

2.25.1 What We Accomplished

This comprehensive assignment demonstrated mastery of statistical analysis through five distinct but interconnected research questions. Each question required different statistical techniques, yet together they form a cohesive exploration of the teacher ratings dataset.

2.25.2 Key Statistical Techniques Mastered

1. Association Analysis (Question 1) Contingency table construction Chi-square test of independence Effect size calculation (Cramér's V) Risk ratios and odds ratios Interpretation of categorical relationships

Real-World Application: Understanding demographic patterns in career progression, equity analysis, policy evaluation

2. Group Comparisons (Question 2) Independent samples t-test Assumption checking (normality, homogeneity) Cohen's d effect size Confidence interval interpretation Mean differences with practical significance

Real-World Application: A/B testing, clinical trials, educational interventions, performance comparisons

3. Visualization Principles (Question 3) Distribution analysis techniques Bin width optimization Kernel density estimation Normality assessment Critical evaluation of visualization choices

Real-World Application: Data journalism, business intelligence dashboards, scientific publications, exploratory data analysis

4. Communication Skills (Question 4) Understanding plot orientations Categorical data visualization Proportion testing and confidence intervals Design choices based on context Effective data presentation

Real-World Application: Report writing, presentation design, stakeholder communication, data storytelling

5. Robust Statistics (Question 5) Median calculation and interpretation Non-parametric testing (Mann-Whitney U) Rank-based methods Outlier-resistant measures Comparative distribution analysis

Real-World Application: Survey analysis, performance evaluation, quality control, economic indicators

2.25.3 Statistical Thinking Skills Developed

Critical Analysis

- **Question formulation:** Converting research questions into testable hypotheses
- **Method selection:** Choosing appropriate statistical tests for data types
- **Assumption checking:** Verifying test prerequisites before application
- **Interpretation:** Drawing valid conclusions from statistical evidence

Computational Proficiency

- **Data manipulation:** Filtering, grouping, aggregating with pandas
- **Statistical computing:** Implementing tests with scipy and statsmodels

- **Visualization:** Creating publication-quality graphics with matplotlib/seaborn
- **Reproducibility:** Documented, replicable analysis workflow

Mathematical Rigor

- **Formal notation:** Using LaTeX/KaTeX for mathematical expressions
 - **Derivations:** Understanding formulas from first principles
 - **Proof concepts:** Following logical progression from assumptions to conclusions
 - **Theoretical foundations:** Connecting practical analysis to statistical theory
-

2.25.4 Broader Lessons

1. Statistical Significance Practical Significance Every analysis included **both**: - P-values for statistical significance - Effect sizes for practical importance - This dual approach is essential for informed decision-making

2. Assumptions Matter We consistently: - Stated assumptions explicitly - Checked assumptions before testing - Considered alternatives when assumptions violated - This ensures valid, reliable conclusions

3. Visualization Choices Impact Understanding Different plots reveal different patterns: - Histograms show frequency - Box plots show outliers - Violin plots show distribution shape - Choosing wisely enhances communication

4. Context is Crucial Numbers alone aren't enough: - We provided background for each question - Connected findings to real-world implications - Considered limitations and confounds - Acknowledged uncertainty appropriately

2.25.5 Technical Skills Portfolio

Python Libraries: - pandas: Data manipulation and aggregation - numpy: Numerical computations - scipy.stats: Hypothesis testing - matplotlib: Core visualization - seaborn: Statistical graphics - statsmodels: Advanced statistical modeling

Statistical Tests: - Chi-square test of independence - Independent samples t-test - Shapiro-Wilk normality test - Levene's test for homogeneity - Mann-Whitney U test - Binomial test for proportions - Anderson-Darling test

Effect Sizes: - Cramér's V (categorical association) - Cohen's d (standardized mean difference) - Rank-biserial correlation (non-parametric) - Risk ratios and odds ratios

Visualizations: - Histograms with KDE overlays - Box plots and violin plots - Bar charts (vertical and horizontal) - Q-Q plots for normality - Cumulative distribution functions - Grouped and stacked charts - Multi-panel layouts

2.25.6 Methodological Best Practices

Reproducible Research Every analysis step is documented:

```
# Clear, commented code  
# Explicit random seeds (when applicable)  
# Version information for libraries  
# Raw data preserved, transformations documented
```

Comprehensive Reporting Each question includes: 1. Research question and context 2. Variable descriptions 3. Hypotheses (null and alternative) 4. Descriptive statistics 5. Assumption checks 6. Test statistics and p-values 7. Effect sizes and confidence intervals 8. Multiple visualizations 9. Interpretation and conclusions

Mathematical Documentation All formulas provided using KaTeX: - Test statistics - Effect size calculations - Confidence interval formulas - Probability distributions - This ensures transparency and learning

2.25.7 Ethical Considerations Revisited

Throughout this analysis, we maintained ethical standards:

Privacy: Dataset is anonymized; no individual identification possible

Honesty: Reported all results, including non-significant findings

Transparency: Showed all steps, assumptions, and limitations

Responsibility: Acknowledged that statistical associations don't prove causation

Fairness: Analyzed demographic differences with sensitivity to equity concerns

2.25.8 Limitations and Future Directions

Current Study Limitations

1. Observational Data

- Cannot establish causation
- Possible confounding variables
- Selection effects may exist

2. Sample Characteristics

- Single institution or limited scope
- Time period may not generalize
- Missing data on some variables

3. Measurement Issues

- Evaluation scores may have bias
- Self-reported data limitations

- Definition of “visible minority” may vary

Future Research Questions

1. Longitudinal Analysis

- Track professors over time
- See how evaluations change with experience
- Examine tenure attainment trajectories

2. Multivariate Models

- Control for multiple factors simultaneously
- Regression analysis with interaction terms
- Path analysis for mediation effects

3. Qualitative Integration

- Combine with interview data
- Understand mechanisms behind patterns
- Contextualize quantitative findings

4. Cross-Institutional Comparison

- Compare patterns across universities
 - Identify institution-level factors
 - Assess generalizability
-

2.25.9 Skills Transferable to Future Work

Research Applications

- Thesis and dissertation analysis
- Journal article preparation
- Grant proposal development
- Conference presentations

Industry Applications

- Business analytics and KPIs
- A/B testing and experimentation
- User experience research
- Market research and segmentation

Data Science Career

- Exploratory data analysis
- Statistical modeling
- Data visualization
- Communication of technical findings

Academic Excellence

- Critical thinking
 - Problem-solving
 - Attention to detail
 - Clear communication
-

2.25.10 Assignment Quality Checklist

Completeness: All 5 questions fully addressed **Correctness:** Statistical methods properly applied **Clarity:** Clear explanations and interpretations **Comprehensiveness:** Multiple approaches per question **Computations:** Accurate calculations with formulas **Code Quality:** Clean, commented, reproducible **Visualizations:** Professional, informative graphics **Mathematical Rigor:** Formal notation and derivations **Context:** Connected to real-world applications **Critique:** Acknowledged limitations and assumptions

2.25.11 Personal Growth

This assignment represents significant progress in:

Technical Skills: - Proficiency with Python data science stack - Mastery of diverse statistical tests - Creation of publication-quality visualizations

Conceptual Understanding: - Deep grasp of hypothesis testing framework - Appreciation for effect sizes - Understanding of assumption checking

Professional Development: - Scientific communication skills - Attention to reproducibility - Ethical awareness in data analysis

Problem-Solving: - Breaking complex questions into steps - Choosing appropriate methods - Interpreting results in context

2.25.12 Closing Thoughts

Statistical analysis is both an **art** and a **science**:

- **Science:** Rigorous methods, mathematical foundations, reproducible procedures
- **Art:** Choosing visualizations, crafting interpretations, telling data stories

This assignment demonstrated that excellent statistical work requires: 1. **Technical skill** - knowing the methods 2. **Statistical thinking** - understanding when and why to apply them 3. **Communication** - explaining findings clearly 4. **Ethics** - maintaining integrity and responsibility 5. **Curiosity** - asking good questions and seeking patterns

These five questions about teacher ratings have provided a comprehensive foundation for future statistical endeavors.

Assignment Complete: Ready for Professional Review

Total Cells: 35+ Lines of Code: 1,000+ Visualizations: 30+ Mathematical Formulas: 100+ Pages (when rendered): 50+

Thank you for this opportunity to demonstrate comprehensive statistical analysis skills.

2.26 References and Resources

Statistical Methods: - Field, A. (2018). *Discovering Statistics Using IBM SPSS Statistics* (5th ed.) - Agresti, A. (2018). *Statistical Methods for the Social Sciences* (5th ed.) - Wilcox, R. R. (2017). *Introduction to Robust Estimation and Hypothesis Testing* (4th ed.)

Python for Data Science: - McKinney, W. (2022). *Python for Data Analysis* (3rd ed.) - VanderPlas, J. (2016). *Python Data Science Handbook* - Géron, A. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (2nd ed.)

Visualization: - Tufte, E. R. (2001). *The Visual Display of Quantitative Information* (2nd ed.) - Wilke, C. O. (2019). *Fundamentals of Data Visualization* - Cairo, A. (2016). *The Truthful Art: Data, Charts, and Maps for Communication*

Documentation: - pandas: <https://pandas.pydata.org/docs/> - matplotlib: <https://matplotlib.org/stable/contents.html> - seaborn: <https://seaborn.pydata.org/> - scipy.stats: <https://docs.scipy.org/doc/scipy/reference/stats.html>

END OF ASSIGNMENT 4